A simple way of looking at these results is to consider the equation

$$Ax_n = \lambda_n x_n = \int_a^b a(t, t')x_n(t')\, dt'$$

from the viewpoint of simple Riemann integration theory. If we subdivide the interval of integration into N parts, each of length l/N, where l is the length of the interval, then our equation can be written as

$$\sum_{\nu=1}^{N} a(t, t_\nu)x_n(t_\nu)\left(\frac{l}{N}\right) = \lambda_n x_n(t) ,$$

where t_ν is some point in the νth interval. If N is large enough this will approximate the original integral arbitrarily closely, as long as the functions involved are continuous. Thus

$$\sum_{\nu=1}^{N} a(t_{\nu'}, t_\nu)x_n(t_\nu)\left(\frac{l}{N}\right) = \lambda_n x_n(t_{\nu'}) .$$

Defining

$$\tilde{M}_{\nu'\nu} \equiv a(t_{\nu'}, t_\nu) , \qquad \tilde{C}_\nu^{(n)} \equiv x_n(t_\nu) ,$$

we get simply

$$\frac{l}{N}\sum_{\nu=1}^{N} \tilde{M}_{\nu'\nu}\tilde{C}_\nu^{(n)} = \lambda_n \tilde{C}_{\nu'}^{(n)} , \tag{9.50}$$

which obviously bears a very close relation to Eq. (9.48). We see that the basic idea behind the practical solution of integral equations is no more complicated than the basic idea involved in evaluating the Riemann integral.

For $||A_1|| \ll 1$, the approximate eigenvalues of A are given by Eq. (9.48) and the related eigenvectors by Eqs. (9.48) and Eq. (9.49). To estimate the size of the error involved in the approximation, we can make use of the perturbation theory results of Eqs. (9.39), (9.40), (9.41), and (9.42). A particularly important point in this connection is the fact that if $x_m^{(0)}$ and $x_n^{(0)}$ are any approximate solutions of the form

$$x_m^{(0)} = \sum_{\nu=1}^{N} C_\nu^{(m)}\phi_\nu^{(N)} , \qquad x_n^{(0)} = \sum_{\nu=1}^{N} C_\nu^{(n)}\phi_\nu^{(N)} , \tag{9.51}$$

then

$$(x_m^{(0)}, A_1 x_n^{(0)}) = (x_m^{(0)}, (A - A_N)x_n^{(0)}) = 0 \tag{9.52}$$

so that the perturbation theory results simplify considerably. Indeed, at first glance it might seem that they simplify too much, namely, to

$$\lambda_n = \lambda_n^{(0)} \qquad \text{and} \qquad x_n = x_n^{(0)} ,$$

since all the terms contributing to $f_n^{(1)}, f_n^{(2)}$, etc., in Eq. (9.40) and to $\lambda_n^{(1)}$, $\lambda_n^{(2)}$, etc., in Eq. (9.42) appear to vanish. However, this is not the case. Vectors of

the form of Eq. (9.51) are *not* the only eigenvectors of A_N. *Any* vector which is orthogonal to all N of the $\phi_n^{(N)}$ is an eigenvector of A_N with eigenvalue zero. There is an infinite number of such vectors, and we have not determined any of them. If we agree to label these starting with index $N+1$, that is, $x_{N+1}^{(0)}$, $x_{N+2}^{(0)}, \cdots$, then the corrections to x_n and λ_n can be put in the form

$$\begin{aligned} x_n &= x_n^{(0)} + \frac{1}{\lambda_n^{(0)}} \sum_{\nu=N+1}^{\infty} (x_\nu^{(0)}, A_1 x_n^{(0)}) x_\nu^{(0)} + \cdots , \\ \lambda_n &= \lambda_n^{(0)} + \frac{1}{\lambda_n^{(0)}} \sum_{\nu=N+1}^{\infty} |(x_\nu^{(0)}, A_1 x_n^{(0)})|^2 + \cdots . \end{aligned} \tag{9.53}$$

Since $A_0 x_\nu^{(0)} = 0$ for $\nu > N$, these can also be written as

$$\begin{aligned} x_n &= x_n^{(0)} + \frac{1}{\lambda_n^{(0)}} \sum_{\nu=N+1}^{\infty} (x_\nu^{(0)}, A x_n^{(0)}) x_\nu^{(0)} + \cdots , \\ \lambda_n &= \lambda_n^{(0)} + \frac{1}{\lambda_n^{(0)}} \sum_{\nu=N+1}^{\infty} |(x_\nu^{(0)}, A x_n^{(0)})|^2 + \cdots . \end{aligned} \tag{9.54}$$

In particular, note that the leading correction to $\lambda_n^{(0)}$, $(x_n^{(0)}, A_1 x_n^{(0)})$, vanishes. Equations (9.53) indicate that the first nonvanishing correction to $\lambda_n^{(0)}$ is always positive if $\lambda_n^{(0)}$ is positive, and that this correction is certainly less than or equal to $||A_1||^2/\lambda_n^{(0)}$. However, $||x_n - x_n^{(0)}|| \leq ||A_1||/\lambda_n^{(0)}$, so we have a better estimate on λ_n than on x_n. In either case, the corrections will grow steadily larger as we get to small eigenvalues.

As a typical example, we consider Schrödinger's integral equation for the bound states of a particle in a field given by the centrally symmetric potential $V(|\mathbf{r}|)$. From symmetry considerations we know that the angular part of the bound-state wave function is given by the spherical harmonic $Y_{lm}(\theta, \phi)$ where l is the angular momentum of the particle in question and m is the projection of the angular momentum on the z-axis. The radial wave functions satisfy Eq. (7.94) which we will reproduce here:

$$\phi_l(r) = \frac{2m\kappa}{\hbar^2} \int_0^\infty j_l(i\kappa r_<) h_l^{(1)}(i\kappa r_>) V(r') \phi_l(r') r'^2 \, dr' , \tag{9.55}$$

where κ is related to the binding energy of the particle by

$$\kappa = \sqrt{-2mE/\hbar^2}$$

and $r_>$ and $r_<$ are respectively the greater and lesser of r and r'. Now, as it stands, the kernel of Eq. (9.55) is not self-adjoint, because

$$k(r, r') \neq k^*(r', r) .$$

However, by using the symmetrization method discussed in Problem 8.5, we can easily bring it to this form. We define

$$\psi_l(r) \equiv r V^{1/2}(r) \phi_l(r) .$$

Then Eq. (9.55) becomes

$$\phi_l(r) = \frac{2m\kappa}{\hbar^2}\int_0^\infty rV^{1/2}(r)j_l(i\kappa r_<)h_l^{(1)}(i\kappa r_>)V^{1/2}(r')r'\phi_l(r')\,dr' . \tag{9.56}$$

The kernel is now the symmetric function

$$k_l(r, r') = \frac{2m\kappa}{\hbar^2} rV^{1/2}(r)j_l(i\kappa r_<)h_l^{(1)}(i\kappa r_>)V^{1/2}(r')r' .$$

This satisfies $k_l(r, r') = k_l^*(r', r)$ since, as the reader can easily show from Eqs. (7.86a) and (7.86c), the product $j_l(i\kappa r_<)h_l^{(1)}(i\kappa r_>)$ is always real.

As an illustration, we consider the familiar Coulomb potential,

$$V(r) = -Ze^2/r$$

with $l=0$ (although it is not difficult to work with other values of l). We choose the Coulomb potential in order to have exact results for comparison; actually, from the numerical point of view a potential like the Yukawa potential would be somewhat easier to handle because of its exponential cut-off.

Now, since by Eqs. (7.86a) and (7.86c),

$$j_0(i\kappa r_<) = \frac{\sinh(\kappa r_<)}{\kappa r_<}, \qquad h_0^{(1)}(i\kappa r_>) = -\frac{e^{-\kappa r_>}}{\kappa r_>},$$

we have

$$k_0(r, r') = \frac{2me^2Z}{\hbar^2\kappa}\left[\frac{\sinh(\kappa r_<)e^{-\kappa r_>}}{r_>^{1/2}r_<^{1/2}}\right], \tag{9.57}$$

and the integral equation becomes

$$\phi_0(r) = \frac{2Z}{\kappa a_0}\int_0^\infty \frac{1}{r_>^{1/2}r_<^{1/2}} e^{-\kappa r_>}\sinh(\kappa r_<)\phi_0(r')\,dr' .$$

The quantity $a_0 \equiv \hbar^2/me^2$ has the dimensions of length and is called the *Bohr radius*.

It is convenient to make the problem dimensionless by defining

$$\rho \equiv r/a_0, \qquad q \equiv \kappa a_0 \equiv [-E/(\tfrac{1}{2}mc^2\alpha^2)]^{1/2}, \tag{9.58}$$

where $\alpha \equiv e^2/\hbar c$ is the dimensionless "fine structure" constant. In these units,

$$\phi_0(\rho) = \frac{2Z}{q}\int_0^\infty \frac{1}{\rho_>^{1/2}\rho_<^{1/2}} e^{-q\rho_>}\sinh(q\rho_<)\phi_0(\rho')\,d\rho' . \tag{9.59}$$

Before trying to solve this equation by finite-rank approximation, we should make certain that it is completely continuous by showing that the kernel is square-integrable. We have

$$I = \int_0^\infty d\rho\int_0^\infty d\rho'\,|k_0(\rho, \rho')|^2 = \frac{4Z^2}{q^2}\int_0^\infty d\rho\int_0^\infty d\rho'\,\frac{e^{-2q\rho_>}[\sinh(q\rho_<)]^2}{\rho_>\rho_<} .$$

Assuming that q is positive, the integrations yield

$$I = \frac{\pi^2 Z^2}{6q^2}, \tag{9.60}$$

so that the kernel is indeed square-integrable.

Now the problem posed by Eq. (9.59) is to find the eigenvalues (binding energies), q, for a given Z. However, the kernel appears to have a rather complicated, nonlinear dependence on q. Actually in the present case, this dependence is illusory, since the change of variables

$$x = q\rho, \qquad x' = q\rho', \tag{9.61}$$

leads to the eigenvalue equation

$$2Z \int_0^\infty \frac{e^{-x_>} \sinh (x_<)}{x_>^{1/2} x_<^{1/2}} \psi_0(x')\, dx' = q\psi_0(x). \tag{9.62}$$

Note that the change of variable of Eq. (9.61) supposes that q is positive, as we have already assumed in showing that the kernel of Eq. (9.59) is square-integrable. However, in most cases—for example, if $V(r) = g^2 e^{-\mu r}/r$—it will be impossible to remove the q-dependence in this way; under these circumstances it is often desirable to regard q as fixed and Z as the quantity to be determined. In other words, we ask what potential "strength" will produce a given binding energy.

We can illustrate the problem of obtaining solutions of an integral equation by solving Eq. (9.62) for the hydrogen atom, where $Z = 1$. If we truncate the kernel for $x > 7$, $x' > 7$, and divide the interval $[0, 7]$ into thirty-five equal parts, then the procedure discussed above in obtaining Eqs. (9.47) and (9.48) leads to the problem of calculating the eigenvalues of a 35×35 matrix. With modern computing machines this is an elementary problem. For the first few eigenvalues, we find that

$$q_1 = 0.994, \qquad q_2 = 0.495, \qquad q_3 = 0.326, \qquad q_4 = 0.233.$$

The exact values may be obtained by using the formula for the bound-state energies of hydrogen

$$E_n = -\tfrac{1}{2} mc^2\alpha^2/n^2.$$

According to Eq. (9.58), this means that

$$q_1 = 1 = 1.000, \quad q_2 = \tfrac{1}{2} = 0.500, \quad q_3 = \tfrac{1}{3} = 0.333, \quad q_4 = \tfrac{1}{4} = 0.250.$$

As suggested by Eq. (9.53), the approximate values are always below the exact values. As the eigenvalues get smaller, the agreement worsens, which is also in accord with Eq. (9.53). Note that

$$\sum_{n=1}^{\infty} q_n^2 = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6},$$

which is equal to the integrated square of the kernel of Eq. (9.62), as may be

seen by looking at Eq. (9.60). This is in agreement with the general result of Eq. (9.15). Since the wave functions of hydrogen are known, we can write the exact solutions to Eq. (9.62). Taking into account the changes of variable of Eqs. (9.58) and (9.61), and the definition of ϕ_0 in terms of ϕ_0, we have for the first three eigenvectors (see, for example, L.I. Schiff, *Quantum Mechanics*. New York: McGraw-Hill, 1969)

$$\psi_{10} = 2\sqrt{x}\, e^{-x},$$
$$\psi_{20} = 2\sqrt{2x}(1 - x)e^{-x},$$
$$\psi_{30} = 2\sqrt{3x}(1 - 2x + \tfrac{2}{3}x^2)e^{-x}.$$

These are all orthogonal and normalized to 1 for convenience.

Figure 9.1 shows the exact ψ_{10} (solid curve) compared with the approximate ψ_{10} (dotted curve). The agreement is very close. Since, strictly speaking, the approximate ψ_{10} consists of a series of flat line segments in each subinterval, we have interpolated by replacing this by a point at the middle of the interval. This is an excellent approximation when the curve varies slowly, and is not so good where the curve varies more rapidly (for example, in Fig. 9.1, near the origin). In such regions of rapid variation, one should choose smaller subintervals to obtain higher accuracy. In the framework we have developed, this is easily done. Figures 9.2 and 9.3 show similar comparisons for the second and third eigenvectors. Note that the deviations from the exact result are a bit more noticeable in these cases, although the agreement is still good.

The calculation would have been simpler had we evaluated the matrix in the spirit of the Riemann integral scheme discussed above. There each matrix element is obtained simply by evaluating the kernel at a point in the xx'-plane. For the sake of comparison, we include in Fig. 9.3 the curve which was obtained by this method for ψ_{30}. The corresponding eigenvalue was $q_3 = 0.353$. It is

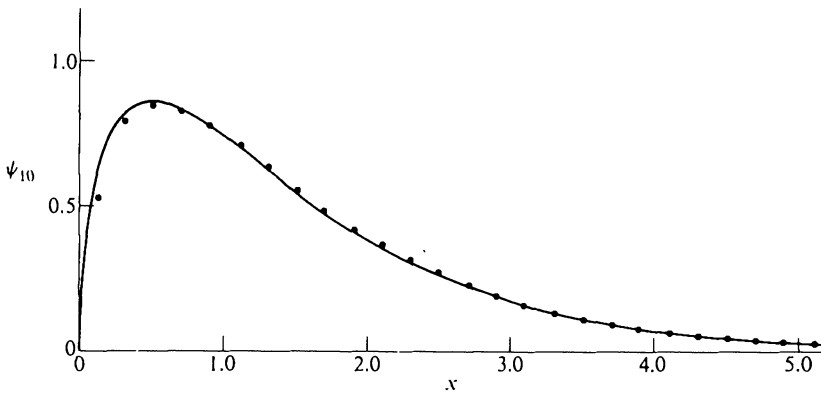

Fig. 9.1 The exact eigenvector, ψ_{10}, solid curve, and the approximate ψ_{10}, dotted curve.

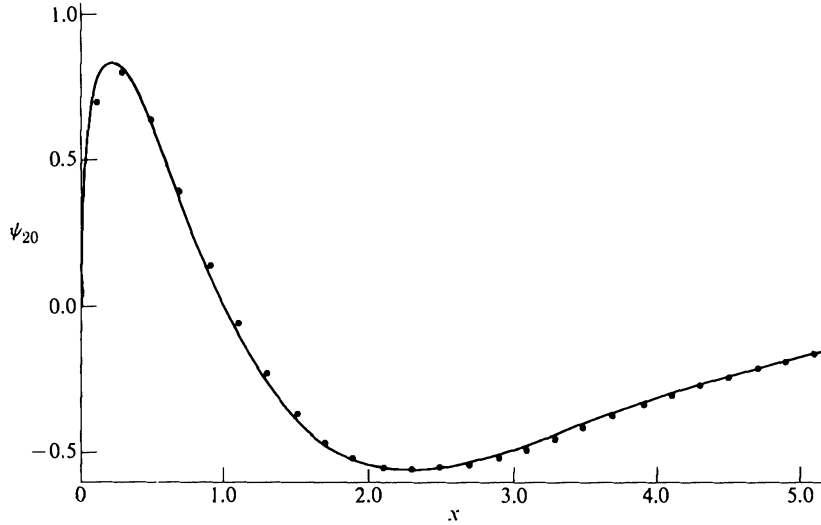

Fig. 9.2 The exact eigenvector, ψ_{20}, solid curve, and the approximate ψ_{20}, dotted curve.

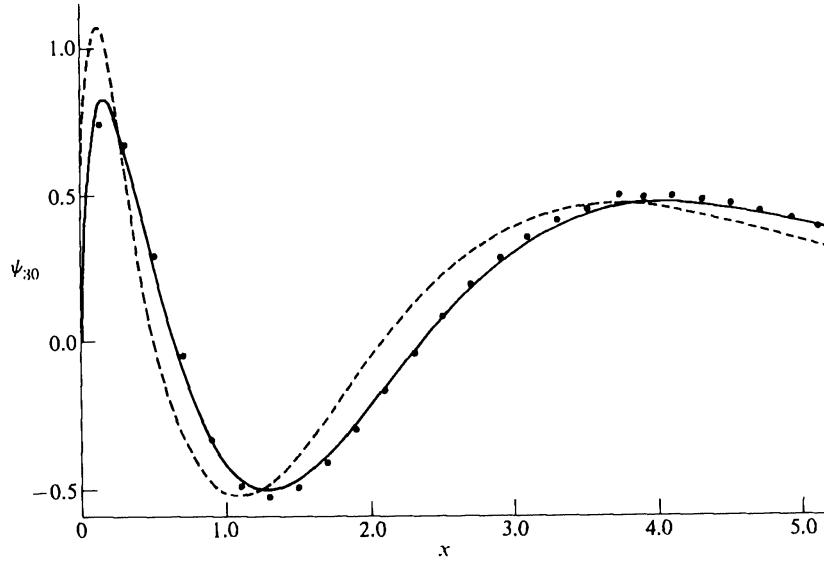

Fig. 9.3 The exact eigenvector, ψ_{30}, solid curve, compared with the approximate ψ_{30} as calculated from Eq. (9.48), dotted curve, and as calculated from Eq. (9.50), dashed curve.

clear that for a coarse subdivision such as the one employed here, the first method is more satisfactory; for sufficiently fine partitions of the interval of integration, the two approaches are equivalent.

9.4 THE FREDHOLM ALTERNATIVE FOR COMPLETELY CONTINUOUS OPERATORS

We now turn our attention to general completely continuous operators. We have less information to work with in this case than we did when considering Hermitian operators. For example, it is no longer possible to guarantee the existence of a complete set of eigenfunctions, and we have no maximum-minimum principles at our disposal. In dealing with non-Hermitian operators, however, we have already made some progress, because we know how to solve linear equations involving either operators with small norm or operators of finite rank. The link between these two types of operators is that any completely continuous transformation of Hilbert space can be written as

$$K = K_F + \Delta\,, \tag{9.63}$$

where K_F is an operator of finite rank and $||\Delta||$ can be made arbitrarily small. Using this basic decomposition, we will show in this section that the Fredholm alternative holds for an arbitrary completely continuous transformation. In doing so, we will also derive several useful results which will enable us to solve problems that arise in practice.

Suppose now that we want to solve

$$f = g + \lambda K f\,, \tag{9.64}$$

or, using Eq. (9.63),

$$(I - \lambda K_F - \lambda\Delta)f = g\,.$$

The problem is to calculate the inverse of $I - \lambda K_F - \lambda\Delta$. We already know how to calculate $(I - \lambda\Delta)^{-1}$ if $||\Delta|| < 1/\lambda$, and how to calculate $(I - \lambda K_F)^{-1}$ when K_F is any kernel of finite rank and λ is not a singular value of K_F. Therefore we would like to write $(I - \lambda K_F - \lambda\Delta)^{-1}$, assuming that it exists, in terms of these known inverses. To do this, we make use of the operator identity (see Problem 3.26)

$$(A - B)^{-1} \equiv A^{-1} + A^{-1}B(A - B)^{-1}\,, \tag{9.65}$$

which is valid if $(A - B)^{-1}$ and A^{-1} both exist. Now for this problem, we identify $(I - \lambda\Delta)$ with A and λK_F with B. Assuming that $||\Delta|| < 1/\lambda$, we know that $(I - \lambda\Delta)^{-1}$ exists and can be calculated by the Neumann series

$$(I - \lambda\Delta)^{-1} = I + \lambda\Delta + \lambda^2\Delta^2 + \lambda^3\Delta^3 + \cdots\,. \tag{9.66}$$

Thus if $(I - \lambda K_F - \lambda\Delta)^{-1}$ exists,

$$(I - \lambda K_F - \lambda\Delta)^{-1} = (I - \lambda\Delta)^{-1} + \lambda(I - \lambda\Delta)^{-1}K_F(I - \lambda K_F - \lambda\Delta)^{-1}\,. \tag{9.67}$$

But if K_F is an operator of finite rank, then so is $(I - \lambda\Delta)^{-1}K_F$, since if K_F is

defined for all $f \in H$ by

$$K_F f \equiv \sum_{i=1}^{N} \phi_i(\psi_i, f) ,$$

then

$$(I - \lambda\Delta)^{-1} K_F f = \sum_{i=1}^{N} (I - \lambda\Delta)^{-1}\phi_i(\psi_i, f) \equiv \sum_{i=1}^{N} \phi_i^{(\Delta)}(\psi_i, f) . \tag{9.68}$$

We have made the definition

$$\phi_i^{(\Delta)} \equiv (I - \lambda\Delta)^{-1}\phi_i = \phi_i + \lambda\Delta\phi_i + \lambda^2\Delta^2\phi_i + \cdots \tag{9.69}$$

Let us define a new operator of finite rank, which depends explicitly on λ:

$$\tilde{K}_F(\lambda) \equiv (I - \lambda\Delta)^{-1} K_F . \tag{9.70}$$

Since $||\Delta|| < 1/\lambda$, the λ-dependence will be of the form of a convergent power series in λ. Thus, adopting the definitions,

$$G_K(\lambda) \equiv (I - \lambda K)^{-1} , \qquad G_\Delta(\lambda) \equiv (I - \lambda\Delta)^{-1} , \tag{9.71}$$

Eq. (9. 67) can be rewritten as

$$G_K(\lambda) = G_\Delta(\lambda) + \lambda\tilde{K}_F(\lambda) G_K(\lambda) . \tag{9.72}$$

In this notation, Eq. (9.70) becomes

$$\tilde{K}_F(\lambda) = G_\Delta(\lambda) K_F . \tag{9.73}$$

Equation (9.72) can be written in the familiar form

$$[I - \lambda\tilde{K}_F(\lambda)] G_K(\lambda) = G_\Delta(\lambda)$$

so all that remains to be done is to calculate the inverse of $[I - \lambda\tilde{K}_F(\lambda)]$, where $\tilde{K}_F$ is a transformation of finite rank. If the inverse exists, then

$$G_K(\lambda) = [I - \lambda\tilde{K}_F(\lambda)]^{-1} G_\Delta(\lambda) \tag{9.74}$$

and our original equation [Eq. (9.64)] is solved by

$$f = G_K(\lambda) g .$$

Equation (9.74) tells us that if $|\lambda| \cdot ||\Delta|| < 1$ (so that $G_\Delta(\lambda)$ exists), then $G_K(\lambda)$ exists if and only if $[I - \lambda\tilde{K}_F(\lambda)]^{-1}$ exists. From a practical point of view, the crucial fact is that Eq. (9.74) enables us to write $G_K(\lambda)$ as the product of two operators, each of which can, in principle, be computed by techniques already discussed: $G_\Delta(\lambda)$ is given by the straightforward iteration of Eq. (9.66), while $[I - \lambda\tilde{K}_F(\lambda)]^{-1}$ can be obtained by the methods of Section 8.6. According to Eqs. (9.68) and (9.70),

$$\tilde{K}_F(\lambda) f = \sum_{i=1}^{N} \phi_i^{(\Delta)}(\psi_i, f)$$

for any $f \in H$, where $\phi_i^{(\Delta)}$ may be written in terms of the original elements of the approximating kernel, K, as

$$\phi_i^{(\Delta)} = G_\Delta(\lambda)\phi_i \, .$$

Thus we can use Eqs. (8.87) and (8.88) to write

$$[I - \lambda \tilde{K}_F(\lambda)]^{-1} = I + \lambda R_{\tilde{K}_F}(\lambda) \, , \tag{9.75}$$

where

$$R_{\tilde{K}_F}(\lambda) f \equiv \sum_{m=1}^{N} \sum_{n=1}^{N} [(I - \lambda A)^{-1}]_{mn} \phi_m^{(\Delta)} (\psi_n, f) \, . \tag{9.76}$$

A is very simple; it is just an $N \times N$ matrix, the mn-element of which is given by

$$A_{mn} = (\psi_m, \phi_n^{(\Delta)}) = (\psi_m, G_\Delta \phi_n) \, . \tag{9.77}$$

According to Eq. (9.69), each element of A can be expressed as a power series in λ, which is convergent if $|\lambda| < ||\Delta||^{-1}$. Thus the original problem can be reduced to one of inverting a finite-dimensional matrix, so in principle we have solved the problem.

It is often possible to use a finite-rank transformation with only a few terms and thereby obtain a Δ whose norm is fairly small. Even with a simple separable kernel of the form

$$k_s(x, y) = \alpha(x)\beta(y) \, ,$$

it is possible in many problems to get a Δ whose norm is much smaller than that of the original K. In specific cases this reduction may be very useful. However, it should be emphasized that no matter how small we make $||\Delta||$, for $|\lambda|$ sufficiently large $|\lambda| \cdot ||\Delta|| \geq 1$, and we can no longer be certain that $(I - \lambda\Delta)^{-1}$ exists.

However, if λ is an arbitrary complex number, then in any circle about the origin the complex λ-plane, there can be only a finite number of points for which $(I - \lambda K)^{-1}$ fails to exist. This is most apparent if we write everything in terms of resolvent operators:

$$G_K(\lambda) \equiv (I - \lambda K)^{-1} \equiv I + \lambda R_K(\lambda) \, ,$$

$$G_\Delta(\lambda) \equiv (I - \lambda \Delta)^{-1} \equiv I + \lambda R_\Delta(\lambda) \, .$$

Using these relations along with Eq. (9.75) in Eq. (9.74), we obtain

$$I + \lambda R_K(\lambda) = (I + \lambda R_{\tilde{K}_F}(\lambda))(I + \lambda R_\Delta(\lambda)) \, ,$$

so

$$R_K(\lambda) = R_\Delta(\lambda) + R_{\tilde{K}_F}(\lambda) \cdot (I + \lambda R_\Delta(\lambda)) \, . \tag{9.78}$$

Note that because of Eqs. (9.69) and (9.77), det $(I - \lambda A)$ is an analytic function of λ which is *not* identically zero (for example, det $(I - \lambda A) = 1$ when $\lambda = 0$). Therefore, if λ_0 is any point in the complex λ-plane for which det

$(I - \lambda_0 A) = 0$, it must be possible to find a neighborhood of λ_0 in which det $(I - \lambda A) \neq 0$ except at λ_0. Otherwise, all the derivatives of det $(I - \lambda A)$ would vanish at $\lambda = \lambda_0$, so the Taylor series about λ_0 would vanish identically. This means, according to the Bolzanno-Weierstrass theorem, that any finite circle in the complex λ-plane contains only a finite number of poles of $R_K(\lambda)$, since according to Eqs. (9.76) and (9.78) zeroes of det $(I - \lambda A)$ correspond to poles of $R_K(\lambda)$. Thus, $R_K(\lambda)$ is a *meromorphic* function of λ.

Another important by-product of breaking K into a transformation of finite rank, K_F, and a transformation with small norm, Δ, is the following result.

Theorem 9.9. The Fredholm alternative holds for any completely continuous transformation of Hilbert space.

Proof. The demonstration of this result is not difficult, but it is a bit long. We shall divide it into a series of lemmas.

Lemma 1. The equations

$$f = g + \lambda K f \tag{9.79}$$

and

$$\tilde{f} = \tilde{g} + \lambda^* K^\dagger \tilde{f}, \tag{9.80}$$

are equivalent, respectively, to the equations

$$f = g_\Delta + \lambda(I + \lambda R_\Delta) K_F f, \tag{9.81}$$

$$\tilde{f} = \tilde{g}_\Delta + \lambda^*(I + \lambda^* R_\Delta^\dagger) K_F^\dagger \tilde{f}, \tag{9.82}$$

where

$$g_\Delta = (I + \lambda R_\Delta) g, \qquad \tilde{g}_\Delta = (I + \lambda^* R_\Delta) \tilde{g},$$

and R_Δ is the resolvent defined above. Δ is an operator which satisfies $||\Delta|| < 1/|\lambda|$, and K_F is an operator of finite rank.

Proof. The result follows immediately from our previous work. We write

$$f = g + \lambda K f = g + \lambda K_F f + \lambda \Delta f,$$

where Δ is such that $||\Delta|| < 1/|\lambda|$. By Theorem 8.4, there is always a K_F for which this will be true. Hence

$$(I - \lambda\Delta) f = g + \lambda K_F f,$$

or, since $(I - \lambda\Delta)^{-1}$ exists and since by definition

$$(I - \lambda\Delta)^{-1} \equiv I + \lambda R_\Delta,$$

we have

$$f = g_\Delta + \lambda(I + \lambda R_\Delta) K_F f$$

with

$$g_\Delta = (I + \lambda R_\Delta) g.$$

An identical argument applied to

$$\tilde{f} = \tilde{g} + \lambda^* K^\dagger \tilde{f}$$

leads to

$$\tilde{f} = \tilde{g}_\Delta + \lambda^*(I + \lambda^* R_\Delta^\dagger) K_F^\dagger \tilde{f} ,$$

as required. By reversing the steps, we immediately see that Eqs. (9.81) and (9.82) imply Eqs. (9,79) and (9.80), respectively, which completes the proof of equivalence.

Lemma 2. Every value of λ is either a regular value or a singular value of the operator K. If it is a regular value, then the equation

$$f = g + \lambda K f \tag{9.83}$$

has a unique solution given by

$$f = (I - \lambda K)^{-1} g ,$$

where $(I - \lambda K)^{-1}$ exists. If λ is a singular value, then Eq. (9.83) has a solution if and only if g is orthogonal to every solution of

$$\lambda^* K^\dagger \tilde{h} = \tilde{h} .$$

Proof. This result follows from Lemma 1 and Theorem 8.1. According to Lemma 1, Eq. (9.83) is equivalent to

$$f = g_\Delta + \lambda(I + \lambda R_\Delta) K_F f \tag{9.84}$$

for some appropriate Δ and K_F. But we have already shown

$$(I + \lambda R_\Delta) K_F$$

to be a transformation of finite rank, since K_F is a transformation of finite rank. Thus, by Theorem 8.1, Eq. (9.84) has a solution if and only if g_Δ is orthogonal to every solution of the homogeneous adjoint equation

$$\lambda^*[(I + \lambda R_\Delta) K_F]^\dagger \tilde{\phi} = \lambda^* K_F^\dagger (I + \lambda R_\Delta)^\dagger \tilde{\phi} = \tilde{\phi} . \tag{9.85}$$

If we call

$$\tilde{h} \equiv (I + \lambda R_\Delta)^\dagger \tilde{\phi} = (I + \lambda^* R_\Delta^\dagger) \tilde{\phi} ,$$

then the requirement that Eq. (9.84) have a solution becomes

$$(g_\Delta, \phi) = 0 = (g_\Delta, [I + \lambda^* R_\Delta^\dagger]^{-1} \tilde{h}) . \tag{9.86}$$

Since $g_\Delta = (I + \lambda R_\Delta) g$, this simply requires that

$$(g, \tilde{h}) = 0 .$$

Now $(I + \lambda^* R_\Delta^\dagger)^{-1}$ exists, and is by definition equal to $(I - \lambda^* \Delta^\dagger)$ (see Lemma 1). Moreover, $K^\dagger = K_F^\dagger + \Delta^\dagger$. Thus, according to Eq. (9.85), $\tilde{h}$ must obey the following equations:

$$\lambda^* K_F^\dagger \tilde{h} = \tilde{\phi} = (I + \lambda^* R_\Delta^\dagger)^{-1} \tilde{h} = (I - \lambda^* \Delta^\dagger) \tilde{h} = \tilde{h} + \lambda^* K_F^\dagger \tilde{h} - \lambda^* K^\dagger \tilde{h} ,$$

or

$$\lambda^* K^\dagger \tilde{h} = \tilde{h} .$$

Hence Eq. (9.84) has a solution if and only if g is orthogonal to all solutions of

$$\lambda^* K^\dagger \tilde{h} = \tilde{h} .$$

By Lemma 1, the same is true for Eq. (9.83), which completes the proof of this lemma.

Corollary. Every value of λ^* is either a regular value or a singular value of the operator $K^\dagger$. If it is a regular value, then the equation

$$\tilde{f} = \tilde{g} + \lambda^* K^\dagger \tilde{f} \tag{9.87}$$

has a unique solution given by

$$\tilde{f} = (I - \lambda^* K^\dagger)^{-1} \tilde{g} ,$$

where $(I - \lambda^* K^\dagger)^{-1}$ exists. If λ is a singular value, then Eq. (9.87) has a solution if and only if $\tilde{g}$ is orthogonal to every solution of $\lambda K h = h$.

Finally, we have

Lemma 3. The equations

$$\lambda K h_i = h_i , \qquad \lambda^* K^\dagger \tilde{h}_i = \tilde{h}_i$$

have the same number of linearly independent solutions.

Proof. According to Lemma 1, these two equations are equivalent respectively to

$$\lambda (I + \lambda R_\Delta) K_F h_i = h_i \tag{9.88}$$

and

$$\lambda^* (I + \lambda^* R_\Delta^\dagger) K_F^\dagger \tilde{h}_i = \tilde{h}_i \tag{9.89}$$

for some Δ and K_F. The last equation can be written as

$$\lambda^* K_F^\dagger (I + \lambda^* R_\Delta^\dagger)[(I + \lambda^* R_\Delta^\dagger)^{-1} \tilde{h}_i] = [(I + \lambda^* R_\Delta^\dagger)^{-1} \tilde{h}_i] , \tag{9.90}$$

since the inverse exists. If we define, as before,

$$\tilde{\phi}_i = (I + \lambda^* R_\Delta^\dagger)^{-1} \tilde{h}_i ,$$

then Eq. (9.90) becomes

$$\lambda^* K_F^\dagger (I + \lambda^* R_\Delta^\dagger) \tilde{\phi}_i = \tilde{\phi}_i ,$$

and Eq. (9.89) becomes

$$\lambda^* [(I + \lambda R_\Delta) K_F]^\dagger \tilde{\phi}_i = \tilde{\phi}_i . \tag{9.91}$$

Now according to the second corollary to Theorem 8.1, Eq. (9.88) and Eq. (9.91) have the same number of linearly independent solutions, $\tilde{h}_i$ and $\tilde{\phi}_i$. But $\tilde{h}_i$ is related to $\tilde{\phi}_i$ by

$$\tilde{h}_i = (I + \lambda^* R_\Delta^\dagger) \tilde{\phi}_i .$$

Since $(I + \lambda^* R_\Delta^\dagger)^{-1}$ exists, the $\tilde{h}_i$ will be linearly independent if the $\tilde{\phi}_i$ are.

Thus Eqs. (9.88) and (9.89) have the same number of linearly independent solutions, and hence the same is true of the equations

$$\lambda K h_i = h_i \qquad \text{and} \qquad \lambda^* K^\dagger \bar{h}_i = \bar{h}_i \,.$$

This completes the proof of Lemma 3 and also of Theorem 9.9.

Note that the Fredholm alternative is exactly the same for completely continuous transformations as for transformations of finite rank. We have tried to emphasize this by retaining the same notation in discussing the two cases (see Section 8.6).

9.5 THE NUMERICAL SOLUTION OF INTEGRAL EQUATIONS

Having assured ourselves that solutions to linear equations with completely continuous kernels exist, it is worthwhile to indicate how one can actually proceed to solve such equations. It is rare that a closed-form solution can be found, so we will concentrate in this section on an approximation scheme which can be refined in principle to arbitrary precision. To do this we will make use of the technique discussed in Section 9.3. However, before applying this method, we must be sure that the eigenvalues of the operators in question behave in a sensible way, i.e., that eigenvalues of approximate kernels come arbitrarily close to the eigenvalues of the kernel which they approximate as the approximation is made arbitrarily fine. For this purpose, let us write

$$(I - \lambda K)^{-1} \equiv (I - \lambda K_F - \lambda \Delta)^{-1}$$

as

$$(I - \lambda K)^{-1} = (I - \lambda K_F)^{-1} + \lambda (I - \lambda K_F)^{-1} \Delta (I - \lambda K)^{-1} \,. \tag{9.92}$$

When we have computed $(I - \lambda K)^{-1}$, the solution to

$$f = g + \lambda K f \tag{9.93}$$

is just

$$f = (I - \lambda K)^{-1} g \,. \tag{9.94}$$

Now we want to approximate K so closely by K_F that the first term in the iterative solution of Eq. (9.92) will provide [via Eq. (9.94)] a satisfactory solution to Eq. (9.93). First, we remark that Eq. (9.92) is valid if and only if $(I - \lambda K)^{-1}$ and $(I - \lambda K_F)^{-1}$ both exist. What we want to show is that if we make $||\Delta||$ sufficiently small, then the values of λ for which $(I - \lambda K)^{-1}$ fails to exist and the values of λ for which $(I - \lambda K_F)^{-1}$ fails to exist get close to each other. (For *Hermitian* operators we have Theorem 9.8 which gives us the desired result in a very precise form.)

For general completely continuous operators, we suppose first that λ is a singular value of K, but that no matter how small $||\Delta||$ becomes $(I - \lambda K_F)^{-1}$ remains bounded, say

$$||(I - \lambda K_F)^{-1}|| < B$$

for *all* Δ. But this assumption leads to a contradiction. For consider the equation

$$X = (I - \lambda K_F)^{-1} + \lambda(I - \lambda K_F)^{-1}\Delta X . \tag{9.95}$$

By familiar arguments, if $|\lambda| \cdot ||(I - \lambda K_F)^{-1}|| \cdot ||\Delta|| < 1$, this equation can be solved by iteration:

$$X = (I - \lambda K_F)^{-1} + \lambda(I - \lambda K_F)^{-1}\Delta(I - \lambda K_F)^{-1} + \cdots . \tag{9.96}$$

But $|\lambda| \cdot ||(I - \lambda K_F)^{-1}|| \cdot ||\Delta|| < 1$ means, by assumption, that $|\lambda| \cdot ||\Delta|| \cdot B < 1$. or

$$||\Delta|| < 1/(|\lambda| \cdot B) .$$

By Theorem 8.4, it is possible to obtain a Δ which satisfies this relation. The operator X is seen to have finite norm; in fact,

$$||X|| \leq \frac{B}{1 - \lambda B||\Delta||} < \infty .$$

But by multiplying the series of Eq. (9.96) on both sides by $(I - \lambda K)$, we find that

$$(I - \lambda K)X = X(I - \lambda K) = I .$$

Thus X is the bounded inverse of $I - \lambda K$. But this is impossible, since λ is assumed to be a singular value of K. Hence as $||\Delta|| \to 0$, $||(I - \lambda K_F)^{-1}|| \to \infty$, that is, λ comes closer and closer to being a singular value of K_F.

Since we know that any given singular value of K will come arbitrarily close to a singular value of K_F for $||\Delta||$ sufficiently small, it remains only to investigate the converse problem: Suppose that λ is *not* a singular value of K; can λ still be a singular value of K_F ? The answer is that if $||\Delta||$ is sufficiently small, then λ is also not a singular value of K_F. Of course, for some finite value of $||\Delta||$ it might happen that λ is a singular value of K_F, but if we keep refining our approximation ($||\Delta|| \to 0$), eventually we can be certain that λ is not a singular value of K_F. To see that this is true, note that Eq. (9.92) implies that if $(I - \lambda K)^{-1}$ exists, then $(I - \lambda K_F)^{-1}$ exists if

$$[I + \lambda\Delta(I - \lambda K)^{-1}]^{-1}$$

exists. In this case,

$$(I - \lambda K_F)^{-1} = (I - \lambda K)^{-1}[I + \lambda\Delta(I - \lambda K)^{-1}]^{-1} .$$

But if $||(I - \lambda K)^{-1}|| < C$, then when $||\Delta|| < 1/|\lambda| \cdot C$ the quantity $[I + \lambda\Delta \cdot (I - \lambda K)^{-1}]^{-1}$ can be computed by iteration. Thus if λ is not a singular value of K for $||\Delta||$ sufficiently small, it must also not be a singular value of K_F. We know, therefore, that if $||\Delta||$ is small enough, and if λ is not a singular value of K, then Eq. (9.92) is well defined and can be solved by iteration:

$$(I - \lambda K)^{-1} = (I - \lambda K_F)^{-1} + \lambda(I - \lambda K_F)^{-1}\Delta(I - \lambda K_F)^{-1} + \cdots . \tag{9.97}$$

For $||\Delta||$ very small compared to $[|\lambda| \cdot ||(I - \lambda K_F)^{-1}||]^{-1}$, we may neglect all

but the first term in Eq. (9.97) and write

$$(I - \lambda K)^{-1} \cong (I - \lambda K_F)^{-1}$$

in the sense of operator norm. Then Eq. (9.94) can be written as

$$f = (I - \lambda K_F)^{-1} g + O(\Delta) ,$$

where $O(\Delta)$ refers to terms which become small in norm as $||\Delta|| \to 0$. Of course, what one means by $||\Delta||$ sufficiently small is clearly, according to Eq. (9.97), dependent on $||(I - \lambda K_F)^{-1}||$. If one is near a singular value of K, and hence of K_F, then $||(I - \lambda K)^{-1}||$ and $||(I - \lambda K_F)^{-1}||$ are large, and $||\Delta||$ must be correspondingly small.

We are now ready to solve linear equations in the same manner employed in the solution of eigenvalue problems involving self-adjoint transformations. Assuming that K is an integral operator, we truncate the kernel $k(x, y)$ in such a way as to ensure the accuracy we desire, and then divide the resulting region on the x(or y)-axis into small subregions. Defining the unnormalized functions $\phi_n^{(N)}(x)$ as in Section 9.3, we obtain a kernel of finite rank, $k_N(x, y)$, which approximates $k(x, y)$ as closely as we please. Proceeding in this way, we find that

$$f = \sum_{n=1}^{N} a_n \phi_n^{(N)} , \tag{9.98}$$

where a_n is given by

$$a_n = \frac{N}{l} \sum_{m=1}^{N} [(I - \lambda A)^{-1}]_{nm} b_m \tag{9.99}$$

and b_m is just

$$b_m = (\phi_m^{(N)}, g) . \tag{9.100}$$

The matrix A is given by

$$A_{\mu\nu} = \frac{N}{l} \int dx \int dy\, k(x, y) \phi_\mu^{(N)}(x) \phi_\nu^{(N)}(y) . \tag{9.101}$$

In these equations l is the length of the interval outside of which we truncate the kernel, and N is the number of subintervals into which we divide the main interval. In obtaining Eq. (9.101), we have assumed that the quantity

$$\left[||g||^2 - \frac{N}{l} \sum_{m=1}^{N} |b_m|^2 \right]^{1/2}$$

is small compared to $||f||$; if the $\phi_n^{(N)}$ formed a complete set, then this quantity would be exactly zero. The reader should derive these results using the method by which we obtained Eqs. (9.48) and (9.49); an expression for f can also be obtained using the Riemann integral approach that led to Eq. (9.50) (see Problem 9.25). With Eqs. (9.98) through (9.101) at our disposal, the problem of solving an integral equation with completely continuous kernel is reduced to a matter of matrix inversion.

As an example of a linear equation with completely continuous kernel, let us consider the Lippmann-Schwinger equation discussed in Chapter 7. Dropping the subscript $(\pm)$ of Eq. (7.68), we write the equation as

$$\psi(\mathbf{r}) = \frac{A}{(2\pi)^{3/2}} e^{i\mathbf{q}_i\cdot\mathbf{r}} - \frac{m}{2\pi\hbar^2}\int \frac{e^{iq_i|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} V(\mathbf{r}')\psi(\mathbf{r}')d^3\mathbf{r}' . \tag{9.102}$$

This is related to the scattering amplitude, $f(\theta, \phi)$, by

$$f^s(\mathbf{q}_f) = -\frac{(2\pi)^{1/2}m}{\hbar^2 A}\int e^{-i\mathbf{q}_f\cdot\mathbf{r}} V(\mathbf{r})\psi(\mathbf{r})\, d^3\mathbf{r} , \tag{9.103}$$

where $\mathbf{q}_f$ is a vector in the direction (θ, ϕ) which is equal in magnitude to $|\mathbf{q}_i|$. As remarked in Section 7.4, f^s is the quantity of real physical interest. Choosing the direction of the incident wave to be the z-direction and using $A = (2\pi)^{3/2}$ for convenience (f^s is clearly independent of A), we get

$$\psi(\mathbf{r}) = e^{iq_i z} - \frac{m}{2\pi\hbar^2}\int \frac{e^{iq_i|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} V(\mathbf{r}')\psi(\mathbf{r}')\, d^3\mathbf{r}' ,$$

$$f^s(\mathbf{q}_f) = -\frac{m}{2\pi\hbar^2}\int e^{-i\mathbf{q}_f\cdot\mathbf{r}} V(\mathbf{r})\psi(\mathbf{r})\, d^3\mathbf{r} .$$

Just as in Section 7.5, we expand everything in terms of Legendre polynomials. We write

$$\psi(\mathbf{r}) = \sum_{l=0}^{\infty} (2l+1)\phi_l(r)P_l(\cos\theta) ,$$

$$f^s(\mathbf{q}_f) = \sum_{l=0}^{\infty} (2l+1)f_l^s(q_i)P_l(\cos\theta_f) ,$$

where we have used the fact that $|\mathbf{q}_f| = q_i$ and have written the scattering angle as θ_f. In this way, we readily obtain a set of equations for the ϕ_l and the f_l^s. We will use the case $l = 0$ as an example. In this case,

$$\phi_0(r) = \frac{\sin(q_i r)}{q_i r} - \frac{2m}{\hbar^2 q_i}\int_0^\infty \frac{\sin(q_i r_<)e^{iq_i r_>}}{r_< r_>} V(r')\phi_0(r')r'^2\, dr' \tag{9.104}$$

$$f_0^s(q_i) = -\frac{2m}{\hbar^2 q_i}\int_0^\infty \sin(q_i r)V(r)\phi_0(r)r\, dr . \tag{9.105}$$

In Eq. (9.104), $r_<$ and $r_>$ denote respectively the smaller of r and r' and the greater of r and r'.

To be specific, let us look at the case of an exponential potential,

$$V(r) = V_0 e^{-\mu r} .$$

Then Eqs. (9.104) and (9.105) become

$$\phi_0(r) = \frac{\sin(q_i r)}{q_i r} - \frac{2mV_0}{\hbar^2 q_i}\int_0^\infty \frac{1}{r}\sin(q_i r_<)e^{iq_i r_>}e^{-\mu r'}r'\phi_0(r')\, dr' , \tag{9.106}$$

$$f_0^s(q_i) = -\frac{2mV_0}{\hbar^2 q_i}\int_0^\infty \sin(q_i r)e^{-\mu r}r\phi_0(r)\, dr . \tag{9.107}$$

As it stands, neither the inhomogeneous term nor the kernel of Eq. (9.106) is square-integrable. However, the symmetrization process which we used in Section 9.3 will bring the equations to the desired form. We define

$$\xi_0(r) \equiv re^{-\mu r/2}\phi_0(r) .$$

Then our equations take the form

$$\xi_0(r) = \frac{1}{q_i} e^{-\mu r/2} \sin (q_i r) - \frac{2mV_0}{\hbar^2 q_i} \int_0^\infty e^{-\mu r/2} \sin (q_i r_<) e^{iq_i r_>} e^{-\mu r'/2} \xi_0(r')\, dr' , \tag{9.108}$$

$$f_0^s(q_i) = -\frac{2mV_0}{\hbar^2 q_i} \int_0^\infty \sin (q_i r) e^{-\mu r/2} \xi_0(r)\, dr . \tag{9.109}$$

It is easily verified that the inhomogeneous term of Eq. (9.108) is square-integrable. What about the kernel

$$k_0(r, r') = -\frac{2mV_0}{\hbar^2 q_i} e^{-\mu(r+r')/2} \sin (q_i r_<) e^{iq_i r_>} ?$$

We find that

$$I(q_i) = \int_0^\infty dr \int_0^\infty dr' |k_0(r, r')|^2 = \frac{2m^2 V_0^2}{\hbar^4 \mu^2 (\mu^2 + q_i^2)} .$$

Then the norm of the operator, K_0, whose kernel is $k_0(r, r')$, satisfies

$$||K_0|| \leq \sqrt{2}\, \frac{mV_0}{\mu\hbar^2} (\mu^2 + q_i^2)^{-1/2} . \tag{9.110}$$

Thus the operator is completely continuous, and its norm tends toward zero as $q_i \to \infty$. Hence for q_i sufficiently large, we can always solve Eq. (9.108) by iteration. One might even hope that it will *always* be possible to solve Eq. (9.108) by iteration. However, for typical situations where an exponential potential is relevant (in nuclear physics, for example), the values of m, μ, and V_0 are such that the estimate on $||K_0||$ reads

$$||K_0|| \leq 2 , \tag{9.111}$$

when $q_i \ll \mu$. We have used

$$1/\mu = 1.4 \times 10^{-13}\,\text{cm} ,$$
$$m = 1.66 \times 10^{-24}\,\text{g} ,$$
$$V_0 = 30\ \text{MeV} = 4.8 \times 10^{-5}\,\text{erg} ,$$

where $1/\mu$ is approximately equal to the range of nuclear forces, m is the mass of the proton (or neutron), V_0 is a reasonable estimate of the strength of nuclear forces, and $\hbar$ is Planck's constant. For $q_i \cong 2\mu$, $||K_0||$ becomes less than 1, but nevertheless for an important range of values of q_i, $||K_0||$ is by no means negligible compared with 1. Of course, Eq. (9.111) is only an *inequality*; $||K_0||$

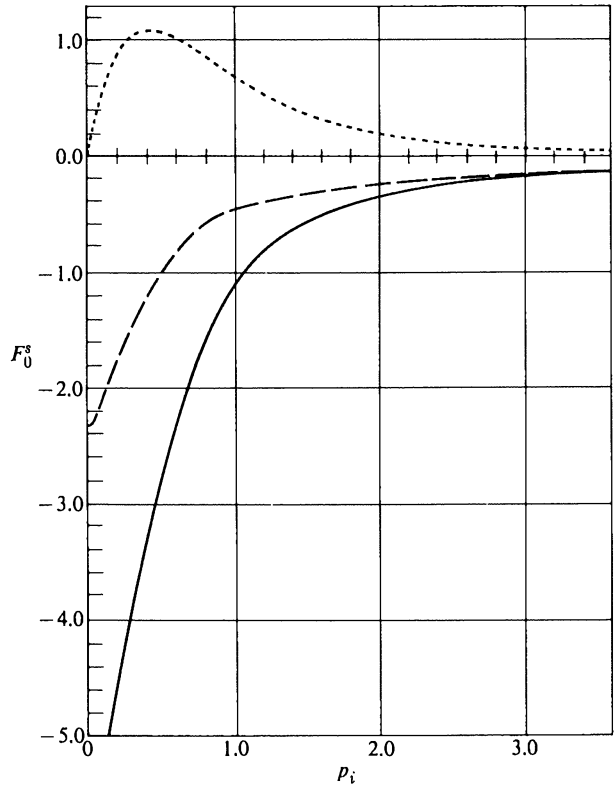

Fig. 9.4 The real part of F_0^s (dashed curve) and the imaginary part of F_0^s (dotted curve) for $l = 0$ scattering by an exponential potential. Included for comparison is the first iterative approximation, F_{0B}^s, which is purely real (solid curve). All the amplitudes are measured in units of the range of the potential $(1/\mu)$, and are plotted as functions of the dimensionless variable p_i, which is essentially the momentum of the scattered particle, $p_i = (2mE/\mu^2\hbar^2)^{1/2}$. The strength of the potential is such that the dimensionless parameter $(mV_0/\mu^2\hbar^2)$ appearing in Eqs. (9.112) and (9.113) is equal to 1.4, as discussed in the text.

might be less than 1, even for $q_i = 0$; however, for the iterative method to be of *practical* use, $||K_0||$ should be *much* less than 1, and the estimate provided by Eq. (9.111) makes that seem unlikely.

Thus we are left with the method which we used in the last section for solving the eigenvalue problem for a Hermitian operator. It is again convenient to scale the distances into appropriate units by defining

$$x \equiv \mu r\,, \qquad p_i = q_i/\mu = (2mE/\mu^2\hbar^2)^{1/2}\,,$$
$$\eta_0 = \mu\xi_0\,, \qquad F_0^s = \mu f_0^s\,.$$

With these changes, Eqs. (9.108) and (9.109) are transformed to

$$\eta_0(x) = \frac{1}{p_i} e^{-x/2} \sin (p_i x) - \frac{2mV_0}{\mu^2\hbar^2 p_i} \int_0^\infty e^{-x/2} \sin (p_i x_<) e^{ip_i x_>} e^{-x'/2} \eta_0(x')\, dx' , \tag{9.112}$$

$$F_0^s(p_i) = -\frac{2mV_0}{\mu^2\hbar^2 p_i} \int_0^\infty \sin (p_i x) e^{-x/2} \eta_0(x)\, dx . \tag{9.113}$$

To solve Eq. (9.112) numerically, we truncate the kernel when x or x' is greater than 7, and then divide the interval [0, 7] into 35 equal parts. Defining $\phi_n^{(N)}$ as usual, we can use Eqs. (9.98) through (9.101) to reduce the problem to one of matrix inversion. Figure 9.4 shows the results for both the real and imaginary parts of F_0^s as a function of p_i for $(mV_0/\mu^2\hbar^2) = 1.4$. This value is obtained from the physically relevant estimates of m, V_0, and μ listed above. For comparison, we also show in Fig. 9.4 the first iterative approximation (Born approximation) to F_0^s, obtained by writing

$$\eta_0^B(x) = \frac{1}{p_i} e^{-x/2} \sin (p_i x) ,$$

$$F_0^{sB} = -2\left(\frac{mV_0}{\mu^2\hbar^2}\right)\frac{1}{p_i} \int_0^\infty \sin (p_i x)\, e^{-x/2} \eta_0^B(x)\, dx = -4\left(\frac{mV_0}{\mu^2\hbar^2}\right)\frac{1}{4p_i^2 + 1} .$$

Just as we expect from Eq. (9.110), for $q_i \leq \mu$ and $p_i \leq 1$ the first iterative approximation to the amplitude differs considerably from the correct

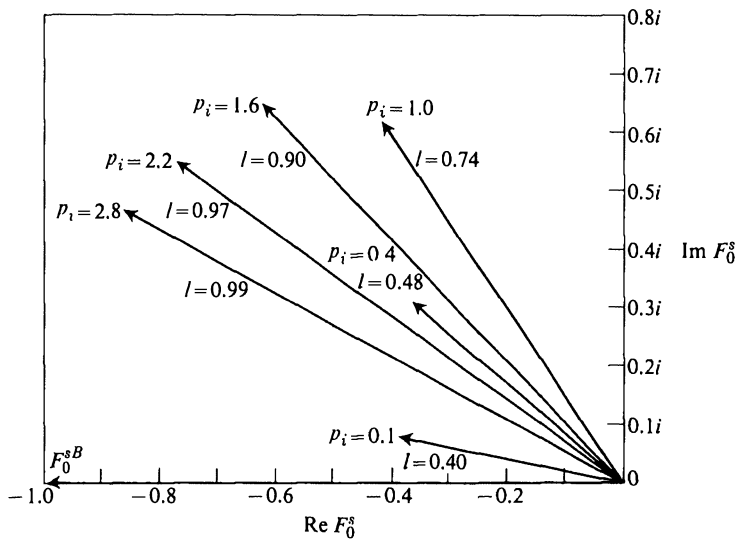

Fig. 9.5 Convergence of F_0^s to the first iterative approximation, as viewed in the complex plane.

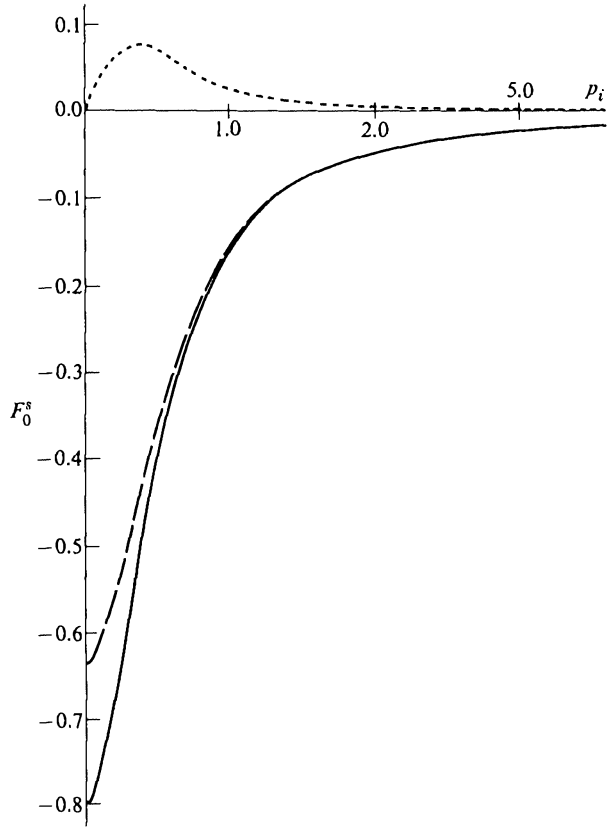

Fig. 9.6 Same as in Figure 9.4, but $(mV_0/\mu^2\hbar^2) = 0.2$.

amplitude. For $p_i = 1$, the real and imaginary parts of F_0^s are of the same order of magnitude, while F_0^{sB} is purely real. However, as p_i becomes larger, the imaginary part of F_0^s drops off more rapidly than the real part, and Fig. 9.4 shows that the real part of F_0^s tends toward F_0^{sB}. However, even for $p_i \cong 3$, the discrepancy is still striking if viewed from the vantage point of Fig. 9.5. In this figure we have divided all amplitudes by $|F_0^{sB}|$ to obtain a reasonable scale and have plotted F_0^s as a "vector" in the complex plane. The "vector" F_0^{sB} is now equal to -1 for all p_i, and we show it as a reference vector along the negative real axis. The tip of each vector is marked by the value of p_i to which it corresponds. We see that the vectors $F_0^s(p_i)$ are tending toward F_0^{sB} as p_i increases, but for $p_i = 2.8$ there is still much improvement to be desired. This is not surprising, for even at $p_i = 2.8$, Eq. (9.110) gives $||K_0|| = 0.66$, so although the relevant norm is less than unity, it is still not particularly small. Along each vector, we have written its length, l, and one is struck by the fact that the length of the complex "vectors" representing $F_0^s(p_i)$ tend very rapidly toward

the length of F_0^{sB}, that is,

$$|F_0^s| \to |F_0^{sB}|$$

as $p_i \to \infty$, much faster than

$$|F_0^s - F_0^{sB}| \to 0 .$$

This means that the most important observable quantity, namely, the cross section, will be given deceptively well by the Born approximation, since it depends only on the *modulus* of F_0^s. However, quantities which depend on the phase of F_0^s will not be given so accurately by the Born approximation.

The iterative approach is much more successful for $(mV_0/\mu^2\hbar^2) = 0.2$. Figure 9.6 shows the real and imaginary part of F_0^s along with the first iterative approximation. Even when $p_i \leq 1$, the imaginary part of F is much smaller than the real part, and by the time $p_i = 1$, F_0^{sB} differs by only three percent from the real part of F_0^s. At $p_i = 3$, the imaginary part of F_0^s is smaller than the real part by more than an order of magnitude, and F_0^{sB} and the real part of F_0^s are virtually identical. Thus, in this case, the iterative approach is simple and accurate. But unfortunately, this will not be the case in many problems of practical interest, and more powerful techniques involving some form of finite-rank approximation must be used.

*9.6 UNITARY TRANSFORMATIONS

Up to now, most of our detailed work has been on completely continuous transformations of Hilbert space, although we managed to put the discussion of iterative techniques in the more general setting of *bounded* transformations. In this section we want to touch on unitary transformations of Hilbert spaces, which are important in mathematical physics because of their connection with the theory of group representations and quantum mechanics, even though they are never completely continuous (see Section 9.1).

At the outset we must make a distinction between unitary and isometric transformations which did not arise in finite-dimensional vector spaces. An isometry is defined in vector spaces of any dimension as a linear transformation U satisfying

$$(Uf, Ug) = (f, g)$$

for all f, g in the vector space. This implies that $U^\dagger$ is a left inverse of U:

$$U^\dagger U = I .$$

Now in a finite-dimensional space, the existence of a left inverse guarantees the existence of a right inverse, but on an infinite-dimensional space this is not always the case. *If* a right inverse also exists, U is said to be unitary. This right inverse must of course be equal to the left inverse, since if $BA = I = AC$, then $BAC = C$, so $B = C$. Thus for *unitary* transformations, we write

$$U^\dagger = U^{-1} .$$

To show that not all isometries are unitary consider a complete orthonormal set in a Hilbert space, H, and define U acting on any $f \in H$ by

$$Uf = \sum_{i=1}^{\infty} \phi_{i+1}(\phi_i, f) .$$

The adjoint transformation $U^\dagger$ is given by

$$U^\dagger g = \sum_{i=1}^{\infty} \phi_i(\phi_{i+1}, g) ,$$

for any $g \in H$. The completeness of the $\{\phi_i\}$ guarantees that

$$(Uf, Ug) = (f, g)$$

for any $f, g \in H$. Thus $U^\dagger$ is a left inverse. Suppose that it were also a right inverse; then we must have

$$(U^\dagger f, U^\dagger g) = (f, UU^\dagger g) = (f, g) ,$$

for any $f, g \in H$, so $U^\dagger$ must be an isometry. But for our $U^\dagger$,

$$(U^\dagger f, U^\dagger g) = \sum_{i=1}^{\infty} (f, \phi_{i+1})(\phi_{i+1}, g) \neq (f, g) ,$$

so $U^\dagger$ is not an isometry and U is therefore *not* unitary. The reader should show that this construction does not produce a nonunitary isometry on a *finite*-dimensional space.

On the space of square-integrable functions, L_2, it is possible to obtain an explicit characterization of all unitary operators. This will lead us to a neater presentation of the Fourier transform, which we have already discussed in an intuitive fashion in Section 5.7. We use the method of Bochner as presented by Riesz and Nagy.

Theorem 9.10. Let U be a unitary transformation of the space of square-integrable functions, L_2. Then for any $f \in L_2$ such that $g = Uf$, there exist two functions $k(\xi, x)$ and $h(\xi, x)$ which satisfy the relations

$$\int_0^\xi g(x)\, dx = \int k^*(\xi, x) f(x)\, dx \tag{9.114a}$$

and

$$\int_0^\xi f(x)\, dx = \int h^*(\xi, x) g(x)\, dx . \tag{9.114b}$$

Proof. We define the function $e_\xi(x)$ which is equal to the sign of x when x lies between zero and ξ and vanishes otherwise. Now for any unitary transformation, let

$$h(\xi, x) \equiv Ue_\xi(x) , \qquad k(\xi, x) \equiv U^{-1}e_\xi(x) = U^\dagger e_\xi(x) .$$

Consider the equation $g = Uf$. We have

$$(e_\xi, g) \equiv (e_\xi, Uf) = (U^\dagger e_\xi, f) = (k(\xi, x), f(x)) .$$

Writing this out explicitly, we get

$$\int_0^\xi g(x)dx = \int k^*(\xi, x)f(x)dx ,$$

making use of the fact that when $e_\xi(x) \not\equiv 0$,

$$e_\xi(x) = \operatorname{sgn}(x)$$

and *not* simply

$$e_\xi(x) = 1 .$$

Similarly,

$$(e_\xi, f) = (e_\xi, U^{-1}g) = (e_\xi, U^\dagger g) = (Ue_\xi, g) = (h(\xi, x), g(x)) ,$$

so that

$$\int_0^\xi f(x)\, dx = \int h^*(\xi, x)g(x)\, dx .$$

This completes the proof of the theorem.

The functions k and h have certain special properties. Suppose that in Eq. (9.114b) we set $f = U^{-1}e_\eta = k(\eta, x)$, and hence $g = Uf = e_\eta$. Thus

$$\int_0^\xi k(\eta, x)\, dx = \int_0^\eta h^*(\xi, x)\, dx . \tag{9.115a}$$

Doing the same in Eq. (9.114a), we find that

$$\int k^*(\xi, x)k(\eta, x)\, dx = \int_0^\xi e_\eta(x)\, dx .$$

Clearly, if ξ and η are of opposite sign, this last integral is zero. If $\xi > 0$, $\eta > 0$, then

$$\int k^*(\xi, x)k(\eta, x)\, dx = \min(\xi, \eta) .$$

If $\xi < 0$, $\eta < 0$, then

$$\int k^*(\xi, x)k(\eta, x)\, dx = -\max(\xi, \eta) = \min(-\xi, -\eta) ,$$
$$= \min(|\xi|, |\eta|) .$$

Hence we have

$$\int k^*(\xi, x)k(\eta, x)\, dx = \begin{cases} \min(|\xi|, |\eta|) & \text{if} \quad \xi\eta > 0 , \\ 0 & \text{if} \quad \xi\eta \leq 0 , \end{cases} \tag{9.115b}$$

and similarly

$$\int h^*(\xi, x)h(\eta, x)\, dx = \begin{cases} \min(|\xi|, |\eta|) & \text{if} \quad \xi\eta > 0 , \\ 0 & \text{if} \quad \xi\eta \leq 0 . \end{cases} \tag{9.115c}$$

The converse of Theorem 9.10 is extremely important from the practical point of view.

Theorem 9.11. Suppose that we are given two functions $k(\xi, x)$ and $h(\xi, x)$ which belong to L_2 for all ξ and which satisfy Eqs. (9.115a), (9.115b), and (9.115c). Then the equations

$$\int_0^\xi g(x)\,dx = \int k^*(\xi, x) f(x)\,dx\,,$$

$$\int_0^\xi f(x)\,dx = \int h^*(\xi, x) g(x)\,dx\,,$$

define a unitary transformation of L_2, U, and its inverse such that $g = Uf$, $f = U^{-1}g$.

Proof. The proof of this result involves a mathematical technique which we have not employed before. The idea is to use the functions k and h to define operators V and U which act on the functions $e_\xi(x)$ introduced in proving Theorem 9.10. We make the definitions

$$Ue_\xi(x) \equiv h(\xi, x)\,, \qquad Ve_\xi(x) \equiv k(\xi, x)\,.$$

For any functions e_ξ, e_η belonging to this class, we have, according to Eq. (9.115a),

$$(Ve_\xi, e_\eta) = (e_\xi, Ue_\eta)\,.$$

Also, Eq. (9.115b) tells us that

$$(Ue_\xi, Ue_\eta) = (e_\xi, e_\eta)\,,$$

and Eq. (9.115c) yields

$$(Ve_\xi, Ve_\eta) = (e_\xi, e_\eta)\,.$$

We would now like to extend the definitions of U and V to a wider class of functions. The first obvious extension is to the class of all elements of L_2 which can be written as *linear combinations* of functions of the type $e_\xi(x)$. For example, we will have

$$U\left[\sum_{n=1}^{N} a_n e_{\xi_n}(x)\right] = \sum_{n=1}^{N} a_n h(\xi_n, x)\,,$$

$$V\left[\sum_{n=1}^{N} a_n e_{\xi_n}(x)\right] = \sum_{n=1}^{N} a_n k(\xi_n, x)\,.$$

By the additivity of the inner product, we have for any ϕ, ψ which belong to this more general class,

$$(V\phi, \psi) = (\phi, U\psi)\,, \qquad (V\phi, V\psi) = (\phi, \psi)\,, \qquad (U\phi, U\psi) = (\phi, \psi)\,.$$

Among the functions in this extended class are the step functions, which take on the values C_n $(n = 1, 2, \cdots, N)$ on the intervals I_n $(n = 1, 2, \cdots, N)$ and are zero elsewhere. But any $f \in L_2$ can be approximated in the mean arbitrarily closely by step functions; that is, we can find a sequence of step functions $\{\phi_n\}$ such that for any $f \in L_2$,

$$||\phi_n - f|| \to 0$$

as $n \to \infty$. Since U is unitary, at least on the class of step functions,

$$||U\phi_m - U\phi_n|| < \epsilon$$

when

$$||\phi_m - \phi_n|| < \epsilon .$$

Therefore, if the ϕ_n converge to some element, f, of L_2, so must the set $\{U\phi_n\}$. We define the function to which this set converges to be Uf. Thus we extend U to the entire space by continuity. It only remains to show that if $f, g \in L_2$, then we still have

$$(Uf, Ug) = (f, g) , \qquad (Vf, Vg) = (f, g) , \qquad (Vf, g) = (f, Ug) .$$

We prove the first of these results and leave the others to the reader. Consider

$$\begin{aligned}(Uf, Ug) - (f, g) \equiv (Uf, Ug) &- (U\phi_n, Ug) + (U\phi_n, Ug) \\ &- (U\phi_n, U\psi_m) + (U\phi_n, U\psi_m) \\ &- (\phi_n, g) + (\phi_n, g) - (f, g)\end{aligned}$$

where $\phi_n \to f$, $\psi_m \to g$. For step functions,

$$(U\phi_n, U\psi_m) = (\phi_n, \psi_m) ,$$

so

$$\begin{aligned}|(Uf, Ug) - (f, g)| &\leq |(Uf, Ug) - (U\phi_n, Ug)| \\ &\qquad + |(U\phi_n, Ug) - (U\phi_n, U\psi_m)| \\ &\qquad + |(\phi_n, \psi_m) - (\phi_n, g)| + |(\phi_n, g) - (f, g)| \\ &\leq |(Uf - U\phi_n, Ug)| + |(U\phi_n, Ug - U\psi_m)| \\ &\qquad + |(\phi_n, \psi_m - g)| + |(\phi_n - f, g)| \\ &\leq ||Uf - U\phi_n|| \cdot ||Ug|| + ||Ug - U\psi_m|| \cdot ||U\phi_n|| \\ &\qquad + ||\psi_m - g|| \cdot ||\phi_n|| \\ &\qquad + ||\phi_n - f|| \cdot ||g|| .\end{aligned}$$

But since $\phi_n \to f$ and $\psi_m \to g$, and by definition $U\phi_n \to Uf$ and $U\psi_m \to Ug$, we can make the right-hand side of this equation as small as we please. Therefore

$$(Uf, Ug) = (f, g) .$$

Similarly,

$$(Vf, Vg) = (f, g) , \qquad (Vf, g) = (f, Ug) ,$$

so that we may conclude that

$$U^\dagger U = I , \qquad V^\dagger V = I , \qquad V = U^\dagger .$$

Hence U has a right inverse and a left inverse:

$$U^{-1} = U^\dagger = V ,$$

that is, U is a unitary transformation whose inverse is given by V. According

to the previous theorem, U and V are represented by

$$\int_0^\xi g(x)\,dx = \int k^*(\xi, x) f(x)\,dx\,,$$

$$\int_0^\xi f(x)\,dx = \int h^*(\xi, x) g(x)\,dx\,,$$

if $g = Uf$. This completes the proof of the theorem.

Note that Eqs. (9.114a) and (9.114b) can be written more conveniently as

$$g(x) = \frac{d}{dx}\int k^*(x, y) f(y)\,dy\,, \tag{9.116a}$$

$$f(x) = \frac{d}{dx}\int h^*(x, y) g(y)\,dy\,. \tag{9.116b}$$

An extremely useful special case of this theorem results when we consider the functions

$$k(x, y) = \frac{1}{\sqrt{2\pi}}\,\frac{e^{-ixy} - 1}{-iy}$$

$$h(x, y) = k^*(x, y)\,.$$

We want to show that these functions satisfy the conditions of Theorem 9.11. Consider first

$$\begin{aligned}\int k^*(\xi, x) k(\eta, x)\,dx &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{(e^{i\xi x} - 1)(e^{-i\eta x} - 1)}{x^2}\,dx \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1}{x^2}[e^{i(\xi-\eta)x} - e^{i\xi x} - e^{-i\eta x} + 1]\,dx \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1}{x^2}[\cos(\xi - \eta)x - \cos\xi x - \cos\eta x + 1]\,dx \\ &\quad + \frac{i}{2\pi}\int_{-\infty}^{\infty} \frac{1}{x^2}[\sin(\xi - \eta)x - \sin\xi x + \sin\eta x]\,dx\,.\end{aligned}$$

The imaginary part vanishes since it is an odd function integrated over a symmetric domain. Note that the integral is well defined, because it converges at $x = \pm\infty$ and is *not* singular at $x = 0$. Thus

$$\begin{aligned}\int k^*(\xi, x) k(\eta, x)\,dx &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1}{x^2}[\cos(\xi - \eta)x - \cos\xi x - \cos\eta x + 1]\,dx \\ &= \frac{1}{\pi}\int_{-\infty}^{\infty} [\sin^2\tfrac{1}{2}|\xi|x + \sin^2\tfrac{1}{2}|\eta|x - \sin^2\tfrac{1}{2}|\xi - \eta|x]\,\frac{dx}{x^2}\,.\end{aligned}$$

Making a change of variable in each term, we get

$$\int k^*(\xi, x) k(\eta, x)\,dx = \frac{1}{2\pi}[|\xi| + |\eta| - |\xi - \eta|]\int_{-\infty}^{\infty} \frac{\sin^2 z}{z^2}\,dz\,.$$

This integral can be readily evaluated by the complex variable methods of Chapter 6; it is equal to π. Thus

$$\int k^*(\xi, x)k(\eta, x)\,dx = \tfrac{1}{2}\,(|\xi| + |\eta| - |\xi - \eta|)$$
$$= \begin{cases} \min\,(|\xi|, |\eta|) & \text{if} \quad \xi\eta > 0\,, \\ 0 & \text{if} \quad \xi\eta \leq 0\,. \end{cases}$$

Likewise,

$$\int h^*(\xi, k)h(\eta, x)\,dx = \begin{cases} \min\,(|\xi|, |\eta|) & \text{if} \quad \eta\xi > 0\,, \\ 0 & \text{if} \quad \eta\xi \leq 0\,. \end{cases}$$

Finally

$$\int_0^\eta k(\xi, x)\,dx = \frac{1}{\sqrt{2\pi}}\int_0^\eta \frac{e^{-i\xi x} - 1}{-ix}\,dx = \frac{1}{\sqrt{2\pi}}\int_0^{\xi\eta} \frac{e^{-iz} - 1}{-iz}\,dz$$

and also

$$\int_0^\xi h^*(\eta, x)\,dx = \frac{1}{\sqrt{2\pi}}\int_0^\xi \frac{e^{-i\eta x} - 1}{-ix}\,dx = \frac{1}{\sqrt{2\pi}}\int_0^{\xi\eta} \frac{e^{-iz} - 1}{-iz}\,dz\,.$$

So

$$\int_0^\xi h^*(\eta, x)\,dx = \int_0^\eta k(\xi, x)\,dx\,.$$

Thus k and h fulfill all the hypotheses of Theorem 9.11, and we have the result:

Corollary. The formulas

$$g(x) = \frac{1}{\sqrt{2\pi}}\frac{d}{dx}\int_{-\infty}^{\infty} \frac{e^{ixy} - 1}{iy} f(y)\,dy \tag{9.117a}$$

and

$$f(x) = \frac{1}{\sqrt{2\pi}}\frac{d}{dx}\int_{-\infty}^{\infty} \frac{e^{-ixy} - 1}{-iy} g(y)\,dy \tag{9.117b}$$

define a unitary transform of L_2 and its inverse.

Note that if we interchange integration and differentiation, we obtain just the classical result of Fourier:

$$g(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{ixy} f(y)\,dy\,, \tag{9.118a}$$

$$f(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-ixy} g(y)\,dy\,. \tag{9.118b}$$

Now we can easily justify this interchange if we interpret properly the infinite integration range. First note that any $f \in L_2$ can be approximated *in norm* arbitrarily closely by a function which vanishes outside a sufficiently large

interval $[-L, L]$, namely, by the function f_L which equals $f(x)$ in $[-L, L]$ and is zero outside. Now let us consider the transform, g_L, of such an approximating function. We have

$$\begin{aligned} g_L(x) &= \frac{1}{\sqrt{2\pi}} \frac{d}{dx} \int_{-L}^{L} \frac{e^{ixy} - 1}{iy} f(y)\, dy \\ &= \frac{1}{\sqrt{2\pi}} \lim_{\epsilon \to 0} \int_{-L}^{L} \frac{e^{i(x+\epsilon)y} - e^{ixy}}{iy\epsilon} f(y)\, dy \\ &= \frac{1}{\sqrt{2\pi}} \lim_{\epsilon \to 0} \int_{-L}^{L} \frac{\sin(\epsilon y/2)}{(\epsilon y/2)} e^{i\epsilon y/2} e^{ixy} f(y)\, dy\,. \end{aligned} \tag{9.119}$$

The function under the integral is less than $f(y)$ in absolute magnitude since

$$\left| \frac{\sin(\epsilon y/2)}{(\epsilon y/2)} e^{i\epsilon y/2} e^{ixy} \right| = \left| \frac{\sin(\epsilon y/2)}{(\epsilon y/2)} \right| \leq 1$$

for all values of ϵy. But on any finite interval, $|f(y)|$ is clearly integrable, because

$$\begin{aligned} \int_{-L}^{L} |f(y)|\, dy &= \int_{-\infty}^{\infty} [e_L(y) - e_{-L}(y)]^* \cdot |f(y)|\, dy \\ &= (e_L, |f|) - (e_{-L}, |f|) < \infty\,, \end{aligned}$$

since e_L, e_{-L}, and $|f|$ all belong to L_2. Therefore, the integrand of Eq. (9.119) is bounded by an integrable function for all values of ϵ, and we can interchange the limit and the integration sign to get

$$g_L(x) = \frac{1}{\sqrt{2\pi}} \int_{-L}^{L} e^{ixy} f(y)\, dy = \frac{1}{\sqrt{2\pi}} \frac{d}{dx} \int_{-\infty}^{\infty} \frac{e^{ixy} - 1}{iy} f_L(y)\, dy\,.$$

Since this last transformation is unitary, $Uf_L \to Uf = g$ as $f_L \to f$ in norm. Thus we may write

$$g(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ixy} f(y)\, dy\,,$$

if we mean by the integral from $-\infty$ to $+\infty$ the limit in norm of the integral from $-L$ to $+L$ as L tends toward infinity. The integration in the inverse transform,

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ixy} g(y)\, dy\,,$$

is to be interpreted in the same way. Thus the Fourier integral gives rise to a unitary transformation of Hilbert space. The transform and its inverse are given by the familiar formulas of Eqs. (9.118a) and (9.118b).

PROBLEMS

1. If A is a completely continuous Hermitian operator with only a finite number of nonzero eigenvalues, show that the eigenvectors of A can be chosen to form a complete orthonormal set.

2. Show that if we modify $k(x, y)$ on a set of measure zero in the plane, then the resulting kernel, $\tilde{k}(x, y)$ has essentially the same eigenvectors and eigenvalues as $k(x, y)$ and the solution to the equation

$$f(x) = g(x) + \lambda \int k(x, y) f(y)\, dy$$

is essentially the same as the solution to

$$\tilde{f}(x) = g(x) + \lambda \int \tilde{k}(x, y)\, \tilde{f}(y)\, dy\,.$$

What is meant by "essentially" in the above statement?

3. Show that if $\{\lambda_n\}$ and $\{x_n(t)\}$ are respectively the eigenvalues and eigenvectors of a completely continuous Hermitian integral operator, then the sum

$$S(t, t') = \sum_{n=1}^{\infty} \lambda_n x_n(t) x_n^*(t')$$

converges in the sense of mean convergence.

4. Prove that if $\{\lambda_n\}$ and $\{x_n(t)\}$ are respectively the eigenvalues and eigenvectors of the Hermitian integral operator, A, with square integrable kernel $a(t, t')$, then

$$\int dt \int dt'\, |a(t, t')|^2 = \sum_{n=1}^{\infty} \lambda_n^2\,.$$

5. Consider the Hermitian integral operator, A, whose square-integrable kernel is $a(t, t')$ and assume for convenience that all the eigenvalues of A are positive.

a) Show that if we write a "trial eigenvector" of A as

$$\phi' = \sum_{n=1}^{N} a_n f_n(t)\,,$$

where $\{f_n(t)\colon\ n = 1, 2, \cdots, \nu\}$ is a given set of functions, then if we extremize $(\phi', A\phi')$ with respect to the coefficients a_n, subject to the constraint $(\phi', \phi') = 1$, the resulting equation for the a_n is

$$Mv = \lambda N v\,. \tag{1}$$

In this equation λ is the Lagrange multiplier which is introduced to handle the constraint, and v is a column vector whose ith component is a_i. M and N are matrices whose ijth elements are

$$M_{ij} = \int f_i^*(t) a(t, t') f_j(t')\, dt\, dt'\,, \qquad i, j = 1, 2, \cdots, \nu\,,$$

$$N_{ij} = \int f_i^*(t) f_j(t)\, dt\,, \qquad i, j = 1, 2, \cdots, \nu\,.$$

b) Show that M and N are Hermitian matrices.

c) Show that the values of λ which satisfy Eq. (1) must be real.

d) Will the eigenvectors corresponding to the various values of λ be orthogonal? Will they satisfy any relationship analogous to the orthogonality relationship?

e) What is the relationship between the largest λ which solves Eq. (1) and the largest eigenvalue of A? We may imagine that if we choose the set $\{f_n(t)\}$ intelligently, it will be possible to arrive at a very good approximation to this largest eigenvalue.

6. a) Let us denote the space of square-integrable functions on the interval [0, 1] by $L_2[0, 1]$. Show that the operator P, defined on $L_2[0, 1]$ by

$$Pf(t) \equiv f(1 - t) ,$$

commutes with the integral operator A, acting on $L_2[0, 1]$, whose kernel is $a(t, t') = t_<(1 - t_>)$.

b) Show that P is self-adjoint and bounded. What is the norm of P?

c) What are the possible eigenvalues of P? Is P completely continuous? If so, why? If not, why not?

d) In accordance with Theorem 4.22 (assuming it to be applicable in this infinite-dimensional case), can you classify the eigenvectors of A [see part (a)] according to which eigenvalue of P they belong to?

7. a) Using the operator P of Problem 9.6, can you give an argument which suggests why the trial function

$$\psi_1' = at(1 - t) + bt^2(1 - t)^2$$

would be a better function for obtaining a good value for the largest eigenvalue of A than the function

$$\psi_2' = at(1 - t) + bt^2(1 - t) .$$

Here A is the Hermitian integral operator on $L_2[0, 1]$ whose kernel is $a(t, t') = t_<(1 - t_>)$. Note that we have already shown in Section 9.2 that $\psi_0' = at(1 - t)$ gives an excellent starting point.

b) Explain why both ψ_1' and ψ_2' must certainly give a value for the largest eigenvalue of A which is at least as close to the true value as the value (computed in Section 9.2) given by ψ_0'.

c) Using the operator P, state directly, without further detailed computation (but making use of the computation in Section 9.2), what the function ψ_2' of part (a) would give as an estimate of the largest eigenvalue of A.

8. a) Show that the function $\psi' = at(1 - t)(1 - 2t)$ is orthogonal to $\sin \pi t$ on the interval [0, 1] so it is an appropriate trial function to use in attempting to find the second largest eigenvalue (and corresponding eigenvector) of the operator A whose kernel is $a(t, t') = t_<(1 - t_>)$ on $L_2[0, 1]$.
[*Hint*: This can be done by brute force integration, but it is much simpler to use arguments involving the operator P of Problem 9.6.]

b) Using the function ψ' given in part (a), obtain an estimate of the second largest eigenvalue of A and compare with the exact value, $\lambda_2 = 1/(4\pi^2)$.

9. Consider the hydrogen atom, consisting of an electron bound to a proton. Neglecting the fact that electrons and protons have spin, one can show on the basis of the nonrelativistic Schrödinger equation,

$$H_0\phi_n = \lambda_n^{(0)}\phi_n ,$$

that there are an infinite number of bound states. Here H_0 is a Hermitian linear operator whose precise form is unimportant for our purpose. The state with lowest energy is a nondegenerate state with eigenvalue $\lambda_1^{(0)} = -13.6$ eV and eigenfunction ϕ_1. The next state with eigenvalue $\lambda_2^{(0)} = -3.4$ eV is fourfold degenerate. We can label the eigenvectors of these four states physically by $\phi_{2s}, \phi_{2p+}, \phi_{2po}$ and ϕ_{2p-}, where the 2 refers to $\lambda_2^{(0)}$, s and p refer to the fact that one state has angular momentum 0 (an "s-state") while the other has angular momentum $\hbar$ (a "p-state")

and the subscripts +, 0, and − refer to the components of angular momentum along the z-axis in the p-state, which can be either $\hbar$, 0 or $-\hbar$. In the first approximation these states are degenerate, but a closer look at the situation reveals that because of electromagnetic self-energy effects, the $2s$-state actually lies above the $2p$-state by an energy L whose value is roughly 4×10^{-6} eV. The states which lie nearest to this closely spaced $2s$-$2p$ cluster are the $3s$, $3p$, and $3d$ states (they are about 2 eV distant).

a) Imagine now that an electric field is applied to the hydrogen atom, so that the states of the system (the atom in the electric field) will be given by the eigenvectors which satisfy

$$H\phi_n = \lambda_n \phi_n ,$$

where $H = H_0 + H_1$, H_0 being the Hermitian linear operator mentioned above, and H_1 being given by $H_1 = eEz$. E is the magnitude of the electric field and e is the electronic charge. We will assume the electric field to be of such a magnitude as to shift the energies of the hydrogen atom by a number of the order of magnitude of L. Assuming the perturbation theory of Section 9.2 to be valid for this problem, explain why in calculating the effect of the field on the $2s$-state we can forget about all other states except the $2p$-states without making a significant error.

b) Show that if we use Eq. (9.34) to calculate the function ψ_{2s} (this is the function which would reduce to ϕ_{2s} if the electric field were turned off) in terms of ϕ_{2s} and ϕ_{2p}, then only the first two terms in Eq. (9.34) are nonvanishing. Show that this would not be the case if we used the analogous Rayleigh-Schrödinger expression of Eq. (9.40). You may use the fact that

$$(\phi_{2s}, H_1\phi_{2p+}) = (\phi_{2s}, H_1\phi_{2p-}) = (\phi_{2p}, H_1\phi_{2p+}) = (\phi_{2p}, H_1\phi_{2p-}) = 0 ,$$

and set the only significant nonvanishing inner product equal to M, that is

$$(\phi_{2s}, H_1\phi_{2po}) = M .$$

c) Using the result of part (b) and utilizing Eqs. (9.33) and (9.34), find ψ_{2s} and λ_{2s} in terms of ϕ_{2s}, ϕ_{2po}, L, and M.

d) Holding E (that is M) fixed, let L become very small. Explain the simple result which you obtain for ψ_{2s} and λ_{2s} in this limit.

10. Consider the quantity

$$\Phi = (f,(A_0 - \lambda_1^{(0)})f) + 2(f(A_1 - \lambda_1^{(1)})x_1^{(0)}) ,$$

where $\lambda_1^{(0)}$ is the *largest* eigenvalue of A_0 with corresponding eigenvector $x_1^{(0)}$. A_1 is a small perturbation on A_0, and $\lambda_1^{(1)} = (x_1^{(0)}, A_1 x_1^{(0)})$. The spectrum of A_0 is assumed to be nondegenerate. A_0 and A_1 are both completely continuous Hermitian operators on a *real* Hilbert space, H. Show that for every $f \in H$,

$$\Phi \leq \lambda_1^{(2)} , \qquad (1)$$

where $\lambda_1^{(2)}$ is given by Eq. (9.42). Show that the equality sign in Eq. (1) holds if and only if f is equal to $f_1^{(1)}$ as given by Eq. (9.40). Note that the assumption that $\lambda_1^{(0)}$ is the largest eigenvalue of A_0 is crucial, and that the assumption of a real Hilbert space is also important. How would one modify Φ if one wanted to extend this result to a complex Hilbert space?

Thus we have a maximum principle for $\lambda_1^{(2)}$ which will enable us to find $\lambda_1^{(2)}$ and $f_1^{(1)}$ by constructing trial functions in the manner indicated in Section 9.2.

Therefore the infinite summations in Eqs. (9.40) and (9.42) can be avoided, at least as far as $f_1^{(1)}$ and $\lambda_1^{(2)}$ are concerned. It can be shown that these sums can be avoided for all $f_i^{(n)}$ and $\lambda_i^{(n)}$ by the use of maximum principles.

11. a) By using Simpson's rule for the evaluation of real integrals, write a matrix expression to solve approximately the equation

$$f(x) = g(x) + \lambda \int_a^b k(x, y) f(y)\, dy\,,$$

where $g(x)$ and $k(x, y)$ are continuous functions. (For a statement and derivation of Simpson's rule, see any book on differential calculus, for example, R. Courant, *Differential and Integral Calculus.*) In particular, show that if the interval $[a, b]$ is broken up by $2n + 1$ equally spaced points $(n > 1)$, $a = x_1$, $x_2, \cdots, x_{2n}$, $b = x_{2n+1}$, the integral equation reduces to

$$a = (I - \lambda MD)^{-1} b\,.$$

Here $a_i = f(x_i)$, $b_i = g(x_i)$, $M_{ij} = k(x_i, x_j)$ and D is a diagonal matrix whose elements are $D_{11} = \frac{1}{3}$, $D_{22} = \frac{4}{3}$, $D_{33} = \frac{2}{3}, \cdots, D_{2n-1,\,2n-1} = \frac{2}{3}$, $D_{2n,\,2n} = \frac{4}{3}$, $D_{2n+1,\,2n+1} = \frac{1}{3}$.

b) Show how to use the same method to treat the homogeneous eigenvalue problem [that is, $g(x) \equiv 0$].

c) Show that if $k(x, y)$ is a real, symmetric kernel, then it is possible to convert the operator eigenvalue equation into a matrix eigenvalue equation involving a single real symmetric matrix.

12. Consider the integral equation

$$f(x) = 1 + 4\int_0^{1/2} \sqrt{1 - xy}\, f(y)\, dy\,. \tag{1}$$

Compare the functions $f_1(x), f_2(x)$, and $f_3(x)$ obtained by making the successive finite-rank approximations:

$$\text{a) } \sqrt{1 - xy} \cong 1\,, \qquad \text{b) } \sqrt{1 - xy} \cong 1 - \tfrac{1}{2}xy\,,$$
$$\text{c) } \sqrt{1 - xy} \cong 1 - \tfrac{1}{2}xy - \tfrac{1}{8}x^2y^2\,,$$

for the kernel of Eq. (1). These are obtained in an obvious manner from the power series for the square root.

13. The purpose of this problem (and the three that follow) is to develop further the Fredholm method of solving a linear integral equation whose kernel is $k(x, y)$. We begin by defining the determinant Δ, a function of $2n$ variables, by

$$\Delta\begin{bmatrix} x_1, x_2, \cdots, x_n \\ y_1, y_2, \cdots, y_n \end{bmatrix} \equiv \begin{vmatrix} k(x_1, y_1) & k(x_1, y_2) & \cdots & k(x_1, y_n) \\ k(x_2, y_1) & k(x_2, y_2) & & k(x_2, y_n) \\ \vdots & & & \vdots \\ k(x_n, y_1) & k(x_n, y_2) & \cdots & k(x_n, y_n) \end{vmatrix}.$$

In what follows, we will assume that $k(x, y)$ is a continuous function for $a \leq x \leq b$ and $a \leq y \leq b$ with $|k(x, y)| \leq M$. Generalizations from this special case to more general situations (e.g., square-integrable kernels on infinite intervals) is straightforward.

a) Consider the infinite series

$$D(\lambda) = 1 - \lambda \int_a^b \Delta\begin{bmatrix} x_1 \\ x_1 \end{bmatrix} dx_1 + \frac{\lambda^2}{2!}\int_a^b\int_a^b \Delta\begin{bmatrix} x_1, x_2 \\ x_1, x_2 \end{bmatrix} dx_1\, dx_2 + \cdots$$
$$+(-1)^n\frac{\lambda^n}{n!}\int_a^b\int_a^b \cdots \int_a^b \Delta\begin{bmatrix} x_1, x_2, \cdots, x_n \\ x_1, x_2, \cdots, x_n \end{bmatrix} dx_1\, dx_2 \cdots dx_n + \cdots .$$

Using Hadamard's inequality (see Section 4.3), prove that the coefficient of $\lambda^n/n!$ is less in magnitude than $M^n n^{n/2}$, and hence show that the series $D(\lambda)$ converges for *all* λ. $D(\lambda)$ is called the *Fredholm determinant.*

b) Consider the infinite series

$$N(x, y, \lambda) = k(x, y) - \lambda \int_a^b \Delta\begin{bmatrix} x, x_1 \\ y, x_1 \end{bmatrix} dx_1 + \frac{\lambda^2}{2}\int_a^b\int_a^b \Delta\begin{bmatrix} x, x_1, x_2 \\ y, x_1, x_2 \end{bmatrix} dx_1\, dx_2 + \cdots$$
$$+ (-1)^n \frac{\lambda^n}{n!}\int_a^b\int_a^b \cdots \int_a^b \Delta\begin{bmatrix} x, x_1, x_2, \cdots, x_n \\ y, x_1, x_2, \cdots, x_n \end{bmatrix} dx_1\, dx_2 \cdots dx_n + \cdots .$$

Using the methods employed in part (a), prove that this series converges for all λ to a continuous function of x and y for $a \le x \le b$ and $a \le y \le b$. (For a discussion of what originally motivated the definition of N and D, see, for example, F. Smithies, *Integral Equations* (Chapter 5).

14. a) Expand the determinant

$$\Delta\begin{bmatrix} x, x_1, x_2, \cdots, x_n \\ y, x_1, x_2, \cdots, x_n \end{bmatrix}$$

by minors about its first row to show that

$$\Delta\begin{bmatrix} x, x_1, x_2, \cdots, x_n \\ y, x_1, x_2, \cdots, x_n \end{bmatrix} = k(x, y)\Delta\begin{bmatrix} x_1, x_2, \cdots, x_n \\ x_1, x_2, \cdots, x_n \end{bmatrix}$$
$$+ \sum_{i=1}^{n} k(x, x_i)\Delta\begin{bmatrix} x_i, x_1, x_2, \cdots, x_{i-1}, x_{i+1}, \cdots, x_n \\ y, x_1, x_2, \cdots, x_{i-1}, x_{i+1}, \cdots, x_n \end{bmatrix}$$

and therefore that

$$\int_a^b\int_a^b \cdots \int_a^b \Delta\begin{bmatrix} x, x_1, x_2, \cdots, x_n \\ y, x_1, x_2, \cdots, x_n \end{bmatrix} dx_1\, dx_2 \cdots dx_n$$
$$= k(x, y)\int_a^b\int_a^b \cdots \int_a^b \Delta\begin{bmatrix} x_1, x_2, \cdots, x_n \\ x_1, x_2, \cdots, x_n \end{bmatrix} dx_1\, dx_2 \cdots dx_n$$
$$- n\int_a^b\int_a^b \cdots \int_a^b k(x, z)\Delta\begin{bmatrix} z, x_1, x_2, \cdots, x_{n-1} \\ y, x_1, x_2, \cdots, x_{n-1} \end{bmatrix} dx_1\, dx_2 \cdots dx_{n-1}\, dz .$$

b) By expanding Δ about the first column, show in the manner of part (a) that

$$\int_a^b\int_a^b \cdots \int_a^b \Delta\begin{bmatrix} x, x_1, x_2, \cdots, x_n \\ y, x_1, x_2, \cdots, x_n \end{bmatrix} dx_1\, dx_2 \cdots dx_n$$
$$= k(x, y)\int_a^b\int_a^b \cdots \int_a^b \Delta\begin{bmatrix} x_1, x_2, \cdots, x_n \\ x_1, x_2, \cdots, x_n \end{bmatrix} dx_1\, dx_2 \cdots dx_n$$
$$- n\int_a^b\int_a^b \cdots \int_a^b \Delta\begin{bmatrix} x, x_1, x_2, \cdots x_{n-1} \\ z, x_1, x_2, \cdots x_{n-1} \end{bmatrix} k(z, y)\, dx_1\, dx_2 \cdots dx_{n-1}\, dz .$$

15. a) Using the result of Problem 9.14(a) in the expression of Problem 9.13(b) for $N(x, y, \lambda)$, show that

$$N(x, y, \lambda) = k(x, y)D(\lambda) + \lambda \int_a^b k(x, z)N(z, y, \lambda)\, dz .$$

Similarly, use the result of Problem 9.14(b) to show that

$$N(x, y, \lambda) = k(x, y)D(\lambda) + \lambda \int_a^b N(x, z, \lambda)k(z, y)\, dz .$$

b) Define

$$R(x, y, \lambda) \equiv \frac{N(x, y, \lambda)}{D(\lambda)} .$$

Show that the two equations of part (a) can be rewritten as

$$(I - \lambda K)(I + \lambda R(\lambda)) = I , \qquad \text{and} \qquad (I + \lambda R(\lambda))(I - \lambda K) = I ,$$

where $R(\lambda)$ is the integral operator whose kernel is $R(x, y, \lambda)$ and K is the integral operator whose kernel is $k(x, y)$. Thus we see that $I + \lambda R(\lambda)$ is the *inverse* of $I - \lambda K$, so that in the language of this chapter, $R(\lambda)$ is the *resolvent* of K. Therefore the equation

$$f = g + \lambda K f$$

can be solved for all values of λ for which $R(\lambda)$ exists by

$$f = g + \lambda R(\lambda) g .$$

In terms of the actual kernels, this may be written as

$$f(x) = g(x) + \lambda \int_a^b R(x, y, \lambda)g(y)\, dy = g(x) + \frac{\lambda}{D(\lambda)} \int_a^b N(x, y, \lambda)g(y)\, dy .$$

16. The only remaining question to settle is: What are the values of λ for which $R(\lambda)$ fails to exist? Clearly, since by Problem 9.13 $N(x, y, \lambda)$ and $D(\lambda)$ are analytic functions of λ, $R(\lambda)$ will fail to exist at any zero of $D(\lambda)$, unless that zero is canceled by a corresponding zero of $N(x, y, \lambda)$. In fact, such a cancellation cannot occur.

a) Using the results of Problem 9.13, show that

$$\frac{dD(\lambda)}{d\lambda} = -\int_a^b N(x, x, \lambda)\, dx .$$

b) Using the result of part (a), show that if $D(\lambda)$ has a zero at $\lambda = \lambda_0$, then $N(x, y, \lambda)$ *cannot* have a zero at this point. Prove this result by contradiction. First, write $D(\lambda) = (\lambda - \lambda_0)d(\lambda)$, where $d(\lambda)$ is analytic; next, assume that $N(x, y, \lambda)$ can also be written in this manner, i.e., as $N(x, y, \lambda) = (\lambda - \lambda_0)n(x, y, \lambda)$, where $n(x, y, \lambda)$ is an analytic function of λ; and finally show that under this assumption the equation of part (a) cannot be satisfied. Thus the zeroes of $D(\lambda)$ are precisely the values of λ for which $R(\lambda)$ does not exist. For these values, the homogeneous equation $f = \lambda K f$ will have a solution.

17. Let K be an integral operator with a continuous kernel and assume that the spectrum of K is nondegenerate. Using the results of Problems 9.15(a) and 9.16(a), show that the residue at a pole, λ_n, of $R(x, y, \lambda)$ is equal to

$$-\frac{\phi_n(x)\phi_n^*(y)}{\int \phi_n(x)\phi_n^*(x)\, dx} ,$$

where ϕ_n satisfies the equation $\lambda_n K\phi_n = \phi_n$, and ψ_n satisfies the adjoint equation $\lambda_n^* K^\dagger \psi_n = \psi_n$. Note that for Hermitian kernels this result is implicit in the eigenfunction expansion of the resolvent given by Eq. (9.22).

18. Show that if $k(x, y) = \phi(x)\psi^*(y)$, then the Fredholm method reduces to

$$D(\lambda) = 1 - \lambda \int \phi(x)\psi^*(x)\, dx\,, \qquad N(x, y, \lambda) = k(x, y) = \phi(x)\psi^*(y)\,,$$

that is, all higher terms in the Fredholm series for $D(\lambda)$ and $N(x, y, \lambda)$ vanish. Show that this leads to a solution of $f = g + \lambda Kf$ in agreement with Eq. (8.57).

19. Assuming that λ is sufficiently small, expand $[D(\lambda)]^{-1}$ in powers of λ and show that through order λ^2, $R(x, y, \lambda)$ is equal to the Neumann series expansion of the resolvent through order λ^2. It can be shown that this equality holds to *all* orders, as we would expect.

20. Suppose that one wished to extend the Fredholm method to square-integrable kernels. The most serious obstacle to doing this is the appearance of terms like

$$I = \int_a^b k(x, x)\, dx$$

in the treatment of Problems 9.13 through 9.16. Even if $k(x, y)$ is square integrable, I may not exist, and hence the method fails. This happens, for example, in the case of kernels of the form $k(\mathbf{r}, \mathbf{r}') = G(\mathbf{r}, \mathbf{r}')V(\mathbf{r})$, where $G(\mathbf{r}, \mathbf{r}')$ is given by Eq. (7.66), even if one makes the appropriate weight-function modification in the manner of Section 8.3.

a) Give an argument which suggests that we can *always* modify $k(x, y)$ in such a way that $k(x, x) = 0$ without altering in any essential way the solutions to

$$f(x) = g(x) + \lambda \int k(x, y)f(y)\, dy\,, \tag{1}$$

where $k(x, y)$ is assumed to be square-integrable.
[*Hint*: See Problem 9.2.]

b) Part (a) seems to suggest that all difficulties with integrals like

$$\int_a^b k(x, x)\, \mathrm{d}x$$

can be removed simply by defining

$$\tilde{D}(\lambda) = 1 - \lambda \int \tilde{\Delta}\begin{bmatrix} x_1 \\ x_1 \end{bmatrix} dx_1 + \frac{\lambda^2}{2!}\int\int \tilde{\Delta}\begin{bmatrix} x_1, x_2 \\ x_1, x_2 \end{bmatrix} dx_1, dx_2 + \cdots$$

$$\tilde{N}(x, y, \lambda) = k(x, y) - \lambda \int \tilde{\Delta}\begin{bmatrix} x, x_1 \\ y, x_1 \end{bmatrix} dx_1 + \frac{\lambda^2}{2!}\int\int \tilde{\Delta}\begin{bmatrix} x, x_1, x_2 \\ y, x_1, x_2 \end{bmatrix} dx_1\, dx_2 + \cdots$$

where $\tilde{\Delta}$ is the same determinant as Δ *except* that whenever an entry of the form $k(x_i, x_i)$ appears in Δ, we replace it by zero to get the equivalent entry in $\tilde{\Delta}$. Thus

$$\Delta\begin{bmatrix} x_1 \\ x_1 \end{bmatrix} = k(x_1, x_1)\,, \qquad \tilde{\Delta}\begin{bmatrix} x_1 \\ x_1 \end{bmatrix} = 0\,,$$

$$\Delta\begin{bmatrix} x_1, x_2 \\ x_1, x_2 \end{bmatrix} = k(x_1, x_1)k(x_2, x_2) - k(x_1, x_2)k(x_2, x_1)\,,$$

$$\tilde{\Delta}\begin{bmatrix} x_1, x_2 \\ x_1, x_2 \end{bmatrix} = -k(x_1, x_2)k(x_2, x_1)\,,$$

and so forth. Letting

$$\tilde{R}(x, y, \lambda) \equiv \frac{\tilde{N}(x, y, \lambda)}{\tilde{D}(\lambda)},$$

and expanding $[\tilde{D}(\lambda)]^{-1}$ (assuming λ is sufficiently small), show that through order λ^2, $\tilde{R}(x, y, \lambda)$ is equal to the Neumann expansion of the resolvent of K. This equality can be shown to hold to all orders. In fact, the modification suggested by this problem is correct, namely, $\tilde{N}(x, y, \lambda)$, $\tilde{D}(\lambda)$, and $\tilde{R}(x, y, \lambda)$ as defined above do indeed provide us with a solution to Eq. (1).

21. Obtain the function $\tilde{D}(\lambda)$, defined in Problem 20, through order λ^2 for the kernel of the integral equation Eq. (9.102) if $V(\mathbf{r}) = g^2 e^{-\mu r}/r$. The variables s, t and u defined in Section 8.3 will be useful. Note that the integral which must be done is very similar to the one we encountered in the evaluation of the norm of this kernel in Section 8.3. Assuming q_i to be real, what is the difference between these two cases?

22. Using Eqs. (9.104) and (9.105), show that if we denote the second iterative approximation to the amplitude by $f_0^{s2B}(q_i)$, then for any real potential $V(r)$ we have

$$Imf_0^{s2B}(q_i) = q_i[f_0^{sB}(q_i)]^2 .$$

23. Calculate the second Born approximation to the s-wave scattering amplitude, $F_0^s(p_i)$, defined by Eqs. (9.112) and (9.113). Compare this result with the curves of Figs. 9.4, 9.5, and 9.6 and make any comments which seem appropriate.

24. It can be shown that the quantity $f_0^s(q_i)$ defined in Section 9.5 can be written as

$$f_0^s(q_i) = \frac{1}{q_i} e^{i\delta_0} \sin \delta_0 ,$$

where δ_0 is a function of q_i and is called the s-wave phase shift. Using Fig. 9.4, obtain a graph of $\delta_0(q_i)$ versus q_i. Are the results obtained for $\delta_0(q_i)$ from the two different curves in Fig. 9.4 consistent? To what extent is δ_0 determined uniquely?

25. a) Obtain Eqs. (9.98) through (9.101) using the methods employed in Section 9.3 to discuss the homogeneous problem.
    b) Derive an equation which "solves" the integral equation $f = g + Kf$ in the spirit of Riemann integration (see Eq. (9.50)).
    c) If you have access to a computer make up a few integral equations and solve them numerically using the results of part (a) or (b). For example, you might consider Eqs. (9.112) and (9.113), but with a potential of the opposite sign (i.e., $mV/\mu^2\hbar^2 = -1.4$). You will find that the scattering amplitude looks rather different than it did in the case of positive V and that there is a striking change in the behavior of δ_0 (see Problem 9.24) as a function of p_i. This change is caused by the existence of an s-wave bound state in the attractive potential (negative V) which is not present in a repulsive potential.

26. Suppose that $\{\phi_n\}$ and $\{\psi_m\}$ are sequences of step functions and that $\phi_n \to f$ and $\psi_m \to g$. Show that if for any step functions ϕ and ψ we have

$$(V\phi, \psi) = (\phi, U\psi) ,$$
$$(V\phi, V\psi) = (\phi, \psi) ,$$

and

$$(U\phi, U\psi) = (\phi, \psi) ,$$

then the sequences $\{V\phi_n\}$ and $\{U\psi_m\}$ converge to some limit in Hilbert space and that if we denote these limits by Vf and Ug respectively, then

$$(Vf, g) = (f, Ug) .$$

This completes the proof of Theorem 9.11.

27. a) Show that if we define

$$\mathscr{H}_n(x) \equiv H_n(x)e^{-x^2/2} ,$$

where $H_n(x)$ is the nth Hermite polynomial which satisfies Eq. (5.105), then the Hermite function $\mathscr{H}_n(x)$ satisfies

$$\mathscr{H}_n''(x) - x^2\mathscr{H}_n(x) + (2n + 1)\mathscr{H}_n(x) = 0 .$$

Note that according to Eq. (5.100), $\mathscr{H}_n(x)$ is orthogonal to $\mathscr{H}_m(x)$ on the interval $[-\infty, \infty]$.

b) Denoting the unitary Fourier transformation by U_F, show that the functions $\mathscr{H}_n(x)$ are eigenfunctions of U_F. Do this by proving that the Fourier transform of $\mathscr{H}_n(x)$, say $\hat{\mathscr{H}}_n(x)$, satisfies the same differential equation as does $\mathscr{H}_n(x)$, so $\hat{\mathscr{H}}_n(x)$ must equal (to within a multiplicative constant) $\mathscr{H}_n(x)$. In operator language, this is just

$$U_F\mathscr{H}_n = \lambda_n\mathscr{H}_n ,$$

where λ_n has not yet been determined.
[*Hint* $x^2e^{ixy} = -(d^2/dy^2)e^{ixy}$; also, integration by parts will be useful.]

c) Show that if $f(x)$ is any square-integrable function, then $U_F^2f(x)=f(-x)$. Hence show that the only possible eigenvalues of U_F are 1, -1, i, and $-i$. There is an infinite number of Hermite functions which belong to each of these four eigenvalues. Give a simple characterization of those Hermite functions which will have real eigenvalues and those which will have imaginary eigenvalues.

FURTHER READINGS

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CHAPTER 10

INTRODUCTION TO GROUP THEORY

INTRODUCTION

Thus far in this book we have discussed objects (vectors) which combine in a way that is essentially *additive*, i.e., they combine commutatively. In investigating the properties of these objects we were led in a natural way to consider *transformations* which changed one vector into another. We observed that these transformations themselves could be considered as constituting a vector space. However, since the transformations took one vector into another, they could also be combined in a way which was essentially *multiplicative* in nature; i.e., the commutative law did not in general hold. In this concluding chapter we want to introduce some of the techniques by which one can study collections of objects which are characterized in terms of their multiplicative properties. Such a collection is called a *group*.

10.1 AN INDUCTIVE APPROACH

Let us first recall the definition of a group given previously in Chapter 3:

Definition 10.1. A *group*, G, is a collection of objects which can be combined via a closed operation, which we will denote by a dot. By a closed operation we mean one such that if $a, b \in G$, then $a \cdot b \in G$. The operation must furthermore satisfy the following three axioms:

1) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associative law)
2) there exists an identity element, e, such that $a \cdot e = a$ for all $a \in G$.
3) for every $a \in G$, there exists an inverse element, a^{-1}, such that $a \cdot a^{-1} = e$.

Note that in this definition we have required that the identity be only a *right* identity and that the inverse be only a *right* inverse. However, because of the other group properties, it follows that e is also a *left* identity ($e \cdot a = a$) and a^{-1} is also a *left* inverse ($a^{-1} \cdot a = e$).

To show this we note that since $a^{-1} \in G$, it must have an inverse, which we denote by $(a^{-1})^{-1}$. According to axiom (3),

$$(a^{-1}) \cdot (a^{-1})^{-1} = e\,. \tag{10.1}$$

But by Axioms (1), (2), and (3) we may write Eq. (10.1) as

$$\begin{aligned} e &= a^{-1} \cdot (a^{-1})^{-1} = (a^{-1} \cdot e) \cdot (a^{-1})^{-1} = [a^{-1} \cdot (a \cdot a^{-1})] \cdot (a^{-1})^{-1} \\ &= [(a^{-1} \cdot a) \cdot a^{-1}] \cdot (a^{-1})^{-1} = (a^{-1} \cdot a) \cdot [a^{-1} \cdot (a^{-1})^{-1}]\,. \end{aligned}$$

Using Eq. (10.1) again, we have

$$e = (a^{-1} \cdot a) \cdot e = a^{-1} \cdot (a \cdot e) .$$

By Axiom (2), $a \cdot e = a$, so finally we obtain

$$e = a^{-1} \cdot a . \tag{10.2}$$

Hence a^{-1} is a left inverse as well as a right inverse. This result contrasts with the situation for general linear operators, where the existence of a right inverse does not necessarily imply the existence of a left inverse.

Using this result, we can immediately see that e must be a left identity. By Axiom (2), e is a right identity, that is,

$$a \cdot e = a ,$$

but according to Eq. (10.2), this is equivalent to

$$a \cdot (a^{-1} \cdot a) = a .$$

Using Axioms (1) and (3), we find that

$$a = a \cdot (a^{-1} \cdot a) = (a \cdot a^{-1}) \cdot a = e \cdot a ,$$

that is, we have

$$a = e \cdot a ,$$

so e is also a left identity.

Definition 10.2. A group, G is said to be *abelian* if $a \cdot b = b \cdot a$ for any $a, b \in G$.

Definition 10.3. A group with n elements is called a group of *order n*.

In the first three sections of this chapter, we shall consider only finite groups ($n < \infty$) in order to fix our thinking on the most basic aspects of group theory. In subsequent sections, we shall have some occasion to mention infinite groups; when we do, we shall always state explicitly that this is the case.

Let us now introduce a few groups, starting with the simplest ones.

Example 10.1. The most uncomplicated group which one can imagine is the one element group: $\{e\}$. It obviously satisfies all the requirements of Definition 10.1. It is not a particularly interesting group.

Example 10.2. Next we have the group with two distinct elements: $\{e, a\}$. Now $a \cdot a$ must belong to this group according to Definition 10.1. Thus we must have either $a \cdot a = a$ or $a \cdot a = e$. The former is immediately ruled out by the fact that since a has an inverse, $a \cdot a = a$ implies that $a = e$. This contradicts the assumption that the group has two distinct elements. Thus $a \cdot a = e$. The integers $\{1, -1\}$ form a group with this structure if they combine via ordinary arithmetical multiplication ($e = 1$, $a = -1$). Also, the integers $\{0, 1\}$ form a group with addition modulo two as the rule of combination: $0 + 0 = 0$, $0 + 1 = 1$, $1 + 0 = 1$, $1 + 1 = 0$. In this case, $e = 0$, $a = 1$. Another structure of this type is the group of permutations of two objects; e is the permutation

which leaves the order of the objects unchanged, and a is the permutation which interchanges them.

This last example leads naturally to the following definition.

Definition 10.4. Two groups are said to be *isomorphic* if there exists a one-to-one operation-preserving correspondence between them. That is, suppose that G has elements $a, b, c, \cdots$, and G' has elements $a', b', c', \cdots$ If $(a \cdot b)' = (a' \times b')$ for all $a, b \in G$ and $a', b' \in G$, then the groups $\{G, \cdot\}$ and $\{G', \times\}$ are said to be isomorphic.

Evidently the three groups mentioned in Example 10.2 are isomorphic to each other: once we know everything about any *one* of them, we know everything about *all* of them.

Example 10.3. Let us now move on to groups with three distinct elements, which we denote by $\{e, a, b\}$. From the axioms we see immediately that $a \cdot b = e$, for if $a \cdot b = a$ or $a \cdot b = b$, then the existence of an inverse would imply $b = e$ in the first case and $a = e$ in the second case, which is contrary to our assumption that this group has three *distinct* elements. What about the product $a \cdot a$? Clearly, $a \cdot a = a$ implies that $a = e$, so either $a \cdot a = e$ or $a \cdot a = b$. But

$$a \cdot a = e \Longrightarrow a \cdot (a \cdot b) = b \Longrightarrow a = b ,$$

since $a \cdot b = e$. But this, too, is impossible because we have assumed three distinct elements. Thus $a \cdot a = b$. Similarly $b \cdot b = a$. Adopting the obvious notation for quantities like $a \cdot a$, we may write our three element group as $\{e, a, a^2\}$, with $a^3 = e$. This is the *only* possibility for the three-element group. The postulates of Definition 10.1 restrict us to just one form. A simple example of this group is the set of three cube roots of unity, under ordinary multiplication of complex numbers.

Clearly, it is always possible to construct a group of the form $\{e, a, a^2, a^3, \cdots, a^{n-1}\}$ for any n; the basic model of such a group is the set of n nth roots of unity. In general, consider an arbitrary group, $\{G, \cdot\}$ and let $a \in G$. Form the sequence $a^0 = e, a, a^2, \cdots, a^i, \cdots$, and let n be the smallest *nonzero* integer such that $a^n = e$. Then the element a is said to be of *order* n (if n is infinite, we say that a is of infinite order). In this case, all the elements $a^0, a^1, a^2, \cdots, a^{n-1}$ are distinct, for if $a^i = a^j$ $(i < n, j < n)$, then $a^{|i-j|} = e$. But since $0 \leq i < n$ and $0 \leq j < n$, $|i - j| < n$, which contradicts our assumption that n was the *smallest* nonzero integer for which $a^n = e$.

Definition 10.5. A group whose elements can be written as $\{e, a, a^2, \cdots, a^{n-1}\}$ is called a *cyclic* group (of order n).

Thus far, all our groups have been of the cyclic type, so it is natural to ask: Are all groups cyclic? The answer is provided by the following example.

Example 10.4. We apply the methods already used to the possible groups of order four, which we denote by $\{e, a, b, c\}$. Let us begin by assuming $a^2 = b$;

then we can write the corresponding group as $\{e, a, a^2, c\}$. By an argument used before, $a \cdot c$ must equal either e or a^2. If $a \cdot c = a^2$, we conclude that $a = c$, which contradicts the assumption of four distinct elements. Thus $a \cdot c = e$, so it begins to look like we are obtaining a cyclic group. We therefore look at a^3. Suppose that $a^3 = e$; then $a^3 = a \cdot c$, so $a^2 = c$, which is impossible. If $a^3 = a$, then $a^2 = e$, which again is impossible. Similarly, if $a^3 = a^2$, $a = e$, which is impossible. Thus $a^3 = c$, and we indeed have found a cyclic group of order four: $\{e, a, a^2, a^3\}$, with $a^4 = e$.

The same result is obtained if we assume that $a^2 = c$, except that it then follows by the same reasoning as above that $b = a^3$. As mentioned above, $a^2 = a$ is not permissible. However, there remains the possibility that $a^2 = e$. Clearly, either $a \cdot b = e$ or $a \cdot b = c$. If $a \cdot b = e$, then $a^2 \cdot b = a$; since $a^2 = e$, we have $a = b$, which is impossible. Thus $a \cdot b = c$. By the same process of elimination, we find that $b \cdot a = c$, $a \cdot c = b$ and $c \cdot a = b$. Since $a \cdot b = c$ and $b \cdot a = c$, we have

$$(b \cdot a) \cdot (a \cdot b) = c \cdot c \Longrightarrow b \cdot a^2 \cdot b = c^2 .$$

But $a^2 = e$, so we have immediately $b^2 = c^2$, which means that c^2 cannot equal b or c. Thus either $c^2 = e$ or $c^2 = a$. We first examine the case $c^2 = a$. Then $a \cdot c^2 = a^2 = e$, but $a \cdot c = b$ so we have $b \cdot c = e$, which gives us explicit expressions for the products of all elements of the group. We summarize by writing

$$\begin{gathered} a^2 = e , \qquad b^2 = c^2 = a , \\ b \cdot c = c \cdot b = e , \\ a \cdot b = b \cdot a = c , \qquad a \cdot c = c \cdot a = b . \end{gathered} \tag{10.3}$$

We now want to know if this group of order four is different from the cyclic group of order four which we have already found several times. A very convenient way of analyzing simple finite groups is by means of a *multiplication table*. We illustrate it by using the *cyclic group* of order four:

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

(10.4)

The meaning of this table is simple: in the ij-box one writes the product of the ith element of the group times the jth element. The reader can check for himself that this table represents accurately the fourth-order cyclic group first

discussed in Example 10.4. Now what about the group whose multiplicative properties are given by Eqs. (10.3)? We have for the multiplication table

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

(10.5)

This looks completely different from the table for the fourth-order cyclic group. However, it may nevertheless be *isomorphic* to the fourth-order cyclic group. This will be the case if we can relabel the elements in such a way that the multiplication table is the same as that of the fourth-order cyclic group. If in Table (10.5) we redefine the elements according to

$$e \to e' , \qquad a \to b' ,$$
$$b \to a' , \qquad c \to c' ,$$

then Table (10.5) becomes

	e'	a'	b'	c'
e'	e'	a'	b'	c'
a'	a'	b'	c'	e'
b'	b'	c'	e'	a'
c'	c'	e'	a'	b'

(10.6)

Since Table (10.6) is the same as Table (10.4), we still have not found a group with a multiplicative structure different from that of the cyclic group of order four.

However, in the class of four-element groups, one possibility remains. We have been forced to conclude that $a^2 = e$, $a \cdot b = c$, $b \cdot a = c$, $a \cdot c = b$, $c \cdot a = b$, and $b^2 = c^2$. When we let $b^2 = c^2 = a$, we were led back to the cyclic group. There remains, however, the choice $b^2 = c^2 = e$. Then

$$a \cdot b = c \Longrightarrow a \cdot b^2 = c \cdot b \Longrightarrow c \cdot b = a ,$$

and similarly

$$b \cdot a = c \Longrightarrow b^2 \cdot a = b \cdot c \Longrightarrow b \cdot c = a .$$

This completes the multiplication table:

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

(10.7)

The reader can check that there is no rearrangement of these elements which will give the Table (10.4). Thus we have finally found a noncyclic group. Note, however, that it is an abelian group; this is reflected in the fact that the array in Table (10.7) is symmetric with respect to the diagonal. This group is called the four-group. A simple realization of the four-group is the set of all transformations of a rectangle which leave the rectangle's orientation in space unchanged (remember that in a rectangle the right and left sides are indistinguishable, as are the top and bottom sides). If the rectangle sits on an xy-plane with its center at the origin, there are four such transformations: (i) the identity transformation, (ii) rotation through 180° about an axis perpendicular to the rectangle and passing through its center, (iii) reflection through the x-axis, and (iv) reflection through the y-axis. Note that we omit rotations through 360°, 540°, etc., since they just duplicate the transformations (i) and (ii) above. Figure 10.1 illustrates the configuration. We leave it to the reader to show that this is indeed a group and that its multiplication table is the same as that given in Table (10.7).

From Table (10.7) it also appears that the elements $\{e, a\}$ of the four-group form a group by themselves ($e \cdot e = e$, $e \cdot a = a$, $a \cdot e = a$, $a \cdot a = e$), as do $\{e, b\}$ and $\{e, c\}$. This leads to the following definition.

Definition 10.6. A set of elements H, which is contained in G, is said to be a *subgroup* of G if

i) the product of any pair of elements in H is in H,
ii) if $a \in H$, then $a^{-1} \in H$.

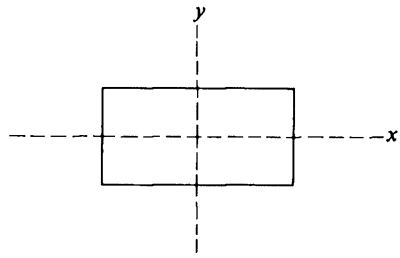

Fig. 10.1

Any subgroup is also a group because its elements are already in G, and hence obey the associative law; moreover, $a, a^{-1} \in H$ implies that $a \cdot a^{-1} = e$ also belongs to H. Clearly any group is a subgroup of itself, and the identity element is a subgroup of any group. As mentioned above, the four-group has $\{e, a\}$ as a subgroup, as well as isomorphic subgroups $\{e, b\}$ and $\{e, c\}$. Clearly, in any group, G, the collection $\{e, a, a^2, \cdots, a^{n-1}\}$, where n is the order of a and $a \in G$, is an abelian subgroup of G.

Another interesting feature of Table (10.7), which it shares with all the other multiplication tables written above, is the fact that each row (or column) contains each element of the group once and only once. This is not accidental; we have in fact the following result.

Theorem 10.1. If G is a group of order n, with elements $e, a_2, a_3, \cdots, a_n$, then every element of G occurs once and only once in the sequence

$$ea_i,\ a_2a_i,\ \cdots,\ a_na_i\ ,$$

for any i, and similarly for

$$a_ie,\ a_ia_2,\ \cdots,\ a_ia_n\ .$$

Note that in the statement of this theorem we have omitted the dot in denoting the product of group elements. We shall do this throughout the chapter, except when this omission might cause confusion.

Proof. If some element occurred twice, we would have $a_\mu a_i = a_\nu a_i$; this would imply $a_\mu = a_\nu$, contrary to our assumption of distinct group elements. Since there are n elements in the sequence and no element occurs twice, each element occurs once and only once. This explains the above mentioned structure of the group multiplication table. This very simple looking theorem is actually of central importance in the proof of most of the results of this chapter.

One could continue inductively, in the manner of this section, developing the groups of order five, order six, etc., but this would not be very instructive. We may remark, however, that not all groups are abelian like the ones which we have discussed so far; at order six one finds the first nonabelian group, the group of permutations of three objects. It should also be emphasized that not all groups have a finite number of elements. For example, the integers under addition ($e = 0$, $n^{-1} = -n$) form an (abelian) group of infinite order; the set of all unitary $n \times n$ matrices is an infinite nonabelian group.

10.2 THE SYMMETRIC GROUPS

By way of further illustrating group structure, we now discuss one of the most important groups of mathematics and physics, the group of permutations on n objects, called the symmetric group, S_n. The structure of the groups S_n is extraordinarily rich, and in this section we will consider only the simplest aspects of these groups.

There are $n!$ permutations of n objects, so S_n is of order $n!$. A typical one of these $n!$ elements will be denoted by

$$p = \begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ m_1 & m_2 & m_3 & \cdots & m_n \end{pmatrix},$$

where the set $\{m_1, m_2, m_3, \cdots, m_n\}$ is some arrangement of the first n integers. This symbol means that 1 is replaced by m_1, 2 by m_2, etc. For example, the permutation

$$p = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 4 & 3 & 2 & 1 \end{pmatrix},$$

acting on the arrangement $\{1\ 2\ 3\ 4\ 5\ 6\}$ produces the arrangement $\{6\ 5\ 4\ 3\ 2\ 1\}$; p acting on $\{2\ 5\ 3\ 4\ 6\ 1\}$ produces $\{5\ 2\ 4\ 3\ 1\ 6\}$. In this example, 1 and 6 go into each other in a closed manner: 1 is replaced by 6, and 6 is replaced by 1. Such a structure is called a *cycle* and will be written simply as (16). Similarly, the above example contains the cycles (25) and (34). We can thus write

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 4 & 3 & 2 & 1 \end{pmatrix} = (16)(25)(34).$$

When we perform such a factorization, we say that we factor the given permutation into *disjoint* cycles, or cycles having no elements in common.

Of course, not all cycles contain only two elements. A cycle containing l elements is called an l-cycle, or a cycle of length l. For example, the permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 4 & 1 & 2 & 5 & 3 \end{pmatrix}$$

can be written as a disjoint product of a three-cycle, a two-cycle, and a one-cycle:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 4 & 1 & 2 & 5 & 3 \end{pmatrix} = (163)(24)(5)\ .$$

The three-cycle, (163), is to be read as "1 is replaced by 6, 6 is replaced by 3 and 3 is replaced by 1." This structure closes on itself just like a two-cycle. Note that

$$(163) = (316) = (631)\ ,$$

but

$$(163) \neq (136)\ .$$

Thus as the name "cycle" suggests, if one writes the elements of a cycle clockwise around a circle, one can start performing the sequence of replacements at *any* element and proceed clockwise back to the starting point. However, if one proceeds in a counterclockwise direction, one gets a different permutation.

The two-cycles, usually referred to as transpositions, are especially important because any cycle (and hence any permutation) can be written as a product of two-cycles (which are not, in general, disjoint). For example,

$$(163) = (13)(16) ,$$

since when (163) acts on say {136}, it produces {613}, and (16) acting on {136} produces {631}, which in turn becomes {613} when acted on by (13). Here we adopt the convention that when a string of cycles is written, the one on the right acts first, and the one on the left acts last. If the cycles are disjoint, it clearly makes no difference in which order they act, but if they are not disjoint, the order is crucial. For example,

$$(163)\{136\} = (13)(16)\{136\} = \{613\} ,$$

whereas

$$(16)(13)\{136\} = \{361\} = (136)\{136\} .$$

Also, the decomposition into two-cycles is *not* unique. In the above case,

$$(163) = (13)(16)$$

and

$$(163) = (316) = (36)(31) .$$

In general, a decomposition of an n-cycle into transpositions can be written as

$$(1\ 2\ 3\ \cdots\ n) = (1n)(1\ n-1)\ \cdots\ (13)(12) ,$$

as may be readily verified. The parity of a permutation is defined to be $(-1)^N$, where N is the number of transpositions in a given permutation. If N is odd, we speak of an odd permutation; if N is even, we speak of an even permutation.

If now by a *product* of permutations, we mean simply the two permutations carried out successively, then obviously the product of two permutations is another permutation. All the other group axioms are trivially satisfied so the collection of all permutations on n objects is a group, and this group will contain $n!$ elements. By our above discussion, it is easily seen that only S_1 and S_2 are abelian groups. S_n $(n \geq 3)$ is nonabelian. For example, (12) and (13) belong to S_3, and $(12)(13) \neq (13)(12)$.

Note that in our convention for writing a permutation, we have quite a bit of freedom in ordering; in the permutation symbol, only the vertical relationship matters, not the horizontal. Thus

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 4 & 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 6 & 5 & 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 4 & 2 & 6 & 3 & 1 & 5 \\ 3 & 5 & 1 & 4 & 6 & 2 \end{pmatrix} .$$

Also, we have used numbers in our symbols merely for convenience (there is an arbitrarily large number of them). One could equally well use apple, pear, orange, $\cdots$, or Greek, Russian, French, $\cdots$, instead of one, two, three, $\cdots$

Example 10.5. The simplest symmetric groups are S_1, which contains the single element e, and S_2, which contains the elements e and (12). More interesting is S_3. Its six elements are

$$e,\ (12),\ (13),\ (23),\ (123),\ (321)\ .$$

As the reader can easily show, the multiplication table for S_3 is

	e	(12)	(13)	(23)	(123)	(321)
e	e	(12)	(13)	(23)	(123)	(321)
(12)	(12)	e	(321)	(123)	(23)	(13)
(13)	(13)	(123)	e	(321)	(12)	(23)
(23)	(23)	(321)	(123)	e	(13)	(12)
(123)	(123)	(13)	(23)	(12)	(321)	e
(321)	(321)	(23)	(12)	(13)	e	(123)

If we make the identifications $e = I$, $(12) = A$, $(13) = B$, $(23) = C$, $(123) = D$, $(321) = F$, then this table is identical to the one given in Table (3.4). Note that this group has quite a few subgroups. Clearly $\{e, (12)\}$ is a subgroup of order two, and $\{e, (13)\}$ and $\{e, (23)\}$ are isomorphic to it. $\{e, (123), (321)\}$ is a subgroup of order three, which is isomorphic to the cyclic group of order three discussed in Example 10.3. Inspection of the multiplication table shows that S_3 is a nonabelian group.

One of the most remarkable facts about S_n is embodied in the following theorem of Cayley.

Theorem 10.2. Every group G of order n is isomorphic to a subgroup of S_n.

Proof. The proof of this result is based on Theorem 10.1. Call the elements of G $a_1, a_2, a_3, \cdots, a_n$. Let a_i be any element of G. Then according to Theorem 10.1, the collection $\{a_ia_1, a_ia_2, \cdots, a_ia_n\}$ is a rearrangement of $\{a_1, a_2, \cdots, a_n\}$ in which every element occurs once and only once. Therefore, let us make the correspondence, which is clearly one-to-one,

$$a_i \to P_{a_i} = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ a_ia_1 & a_ia_2 & \cdots & a_ia_n \end{pmatrix}.$$

Similarly, for $a_j \in G$,

$$a_j \to P_{a_j} = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ a_ja_1 & a_ja_2 & \cdots & a_ja_n \end{pmatrix}.$$

For the element of G which is given by the product a_ia_j, we will have

$$a_ia_j \to P_{a_ia_j} = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ a_ia_ja_1 & a_ia_ja_2 & \cdots & a_ia_ja_n \end{pmatrix}.$$

The crucial point in demonstrating that we have an isomorphism is to show that $P_{a_i}P_{a_j} = P_{a_ia_j}$. This result is easily obtained—it is merely an exercise in notation. The reader should keep this in mind as he plows through the manipulations which follow.

We have already remarked that the horizontal ordering of permutation symbols is unimportant. For example,

$$\begin{pmatrix} a_1 & a_2 & a_3 & \cdots & a_n \\ a_ia_1 & a_ia_2 & a_ia_3 & \cdots & a_ia_n \end{pmatrix} = \begin{pmatrix} a_3 & a_n & a_1 & \cdots & a_2 \\ a_ia_3 & a_ia_n & a_ia_1 & \cdots & a_ia_2 \end{pmatrix}.$$

In other words, we can rearrange the top row in any way we please provided we make a similar rearrangement of the bottom row. Therefore, since $\{a_ja_1, a_ja_2, \cdots, a_ja_n\}$ is a rearrangement of $\{a_1, a_2, \cdots, a_n\}$, we may write

$$P_{a_i} = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ a_ia_1 & a_ia_2 & \cdots & a_ia_n \end{pmatrix} = \begin{pmatrix} a_ja_1 & a_ja_2 & \cdots & a_ja_n \\ a_i(a_ja_1) & a_i(a_ja_2) & \cdots & a_i(a_ja_n) \end{pmatrix}$$
$$= \begin{pmatrix} a_ja_1 & a_ja_2 & \cdots & a_ja_n \\ a_ia_ja_1 & a_ia_ja_2 & \cdots & a_ia_ja_n \end{pmatrix}.$$

Thus

$$P_{a_i}P_{a_j} = \begin{pmatrix} a_ja_1 & a_ja_2 & \cdots & a_ja_n \\ a_ia_ja_1 & a_ia_ja_2 & \cdots & a_ia_ja_n \end{pmatrix}\begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ a_ja_1 & a_ja_2 & \cdots & a_ja_n \end{pmatrix}$$
$$= \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ a_ia_ja_1 & a_ia_ja_2 & \cdots & a_ia_ja_n \end{pmatrix},$$

since the multiplication can now be determined simply by inspection. Thus

$$P_{a_i}P_{a_j} = P_{a_ia_j},$$

and the proof of isomorphism is complete.

This result encompasses the remark made in Example 10.5, where it was noted that S_3 has a subgroup $\{e, (123), (321)\}$ which is isomorphic to the group of order three. Regarding the groups of order four, we take first the four-group. Using the multiplication table of Table (10.7), we find immediately the correspondence

$$P_e = \begin{pmatrix} e & a & b & c \\ e & a & b & c \end{pmatrix}, \qquad P_a = \begin{pmatrix} e & a & b & c \\ a & e & c & b \end{pmatrix},$$
$$P_b = \begin{pmatrix} e & a & b & c \\ b & c & e & a \end{pmatrix}, \qquad P_c = \begin{pmatrix} e & a & b & c \\ c & b & a & e \end{pmatrix},$$

by using the rule of correspondence of Cayley's theorem. In the compact cycle notation, these can be written as

$$P_e = e, \qquad P_a = (ea)(bc), \qquad P_b = (eb)(ac), \qquad P_c = (ec)(ab).$$

It is a simple matter to check that the multiplication table of Table (10.7) is obeyed.

In the case of the cyclic group of order four, we find, using Table (10.4),

$$P_e = \begin{pmatrix} e & a & b & c \\ e & a & b & c \end{pmatrix}, \qquad P_a = \begin{pmatrix} e & a & b & c \\ a & b & c & e \end{pmatrix},$$

$$P_b = \begin{pmatrix} e & a & b & c \\ b & c & e & a \end{pmatrix}, \qquad P_c = \begin{pmatrix} e & a & b & c \\ c & e & a & b \end{pmatrix}.$$

As above, this can be written as

$$P_e = e\,, \qquad P_a = (eabc)\,, \qquad P_b = (eb)(ac)\,, \qquad P_c = (ecba)\,.$$

This case gives us our first example of a four-cycle.

In looking at these two illustrations, one notes that except for the identity, all the permutations leave no symbol unchanged. A subgroup of S_n with this property is called a *regular subgroup* or a subgroup of regular permutations. It is clear from the construction used in proving Theorem 10.2 that every group of order n is isomorphic to a regular subgroup of S_n.

We close this section with a few results about the regular subgroups of S_n.

Lemma. In a regular subgroup, no two elements take a given symbol into the same symbol.

Proof. Suppose that p_1 and p_2 ($p_1 \neq p_2$) belong to a regular subgroup and that both p_1 and p_2 take a into b. But then $p_1 p_2^{-1}$ ($\neq e$) leaves b unchanged. However, $p_1 p_2^{-1}$ also belongs to the regular subgroup (since the subgroup is closed), which contradicts the assumption that the subgroup is regular and hence that every element (except the identity) changes *all* the symbols.

Lemma. In a regular subgroup, if we decompose a given permutation into disjoint cycles, then each cycle must have the same length. For example, in S_5, (12) (345) could not belong to a regular subgroup.

Proof. Suppose that p is an element of a regular subgroup which can be decomposed into two cycles of length l_1 and l_2 with $l_1 < l_2$. Now since a cycle of length l must satisfy $(a_1 a_2 \cdots a_l)^l = e$, p^{l_1} leaves all the symbols contained in the first cycle unchanged. However, p^{l_1} must change some of the symbols contained in the second cycle (all of them, in fact), since if we have a cycle of length $\lambda > l'$, then $(a_1 a_2 \cdots a_\lambda)^{l'} \neq e$. Hence p^{l_1}, which is a permutation belonging to the subgroup containing p, changes some symbols, but not others. This contradicts our assumption that p belongs to a regular subgroup.

This lemma is illustrated by the results just obtained for the regular subgroups of S_4 corresponding to the four-group and to the cyclic group of order four. Using this lemma, we can also prove the following very powerful theorem.

Theorem 10.3. Every group $\{G, \cdot\}$ of order n is cyclic if n is a prime number.

Proof. By the previous lemma, the subgroup of S_n to which G is isomorphic must contain elements which are either a product of n one-cycles or just one n-cycle, since n is prime. The identity is a product of n one-cycles, so G consists of an identity, plus $n - 1$ elements which correspond isomorphically to n-cycles. Call these elements $a_2, a_3, \cdots, a_n$. Now since a_i corresponds to an n-cycle, each element of the sequence $\{a_i, a_i^2, a_i^3, \cdots, a_i^{n-1}\}$ is distinct (note that $a_i^n = e$), since if the elements were not all distinct, say $a_i^\mu = a_i^\nu$, then we would have $a_i^m = e$ for $m < n$, which is impossible because a_i corresponds to an n-cycle. Thus the distinct elements $\{e, a_i, a_i^2, \cdots, a_i^{n-1}\}$ exhaust all elements of the group. That is, we have a cyclic group of order n. Clearly this argument works for any element a_i.

For example, the group of order three is isomorphic to the subgroup $\{e, (123), (321)\}$ of S_3. It is easily checked that

$$(123)^1 = (123)\,, \qquad (123)^2 = (321)\,, \qquad (123)^3 = e\,,$$

so we could write our group as

$$\{e, (123), (123)^2\}\,.$$

But also,

$$(321)^1 = (321)\,, \qquad (321)^2 = (123)\,, \qquad (321)^3 = e\,,$$

so we could equally well reorder the group and write it as

$$\{e, (321), (321)^2\}\,.$$

Theorem 10.3 tells us, among other things, that although we might be tempted to imagine that for n large the number of groups of order n is also large, this is in fact not the case. There is only one group of order 97, namely, the cyclic group.

10.3 COSETS, CLASSES, AND INVARIANT SUBGROUPS

Having discussed in some detail a special category of finite groups, let us now look at some of the most important general properties which are common to all groups.

Definition 10.7. Let A be a subgroup of G, with elements $e, a_2, a_3, \cdots, a_m$ ($m \leq n$, where n is the order of G) and let $b \in G$, but $b \notin A$. Then the m distinct elements $be = b, ba_2, ba_3, \cdots, ba_m$ form a *left coset* of A, which we denote symbolically by bA. Similarly, $b, a_2b, a_3b, \cdots, a_mb$ form a right coset, Ab.

The notation bA is just a shorthand for the collection of objects formed by multiplying every element of A on the left by b. Note that a coset is *not* a subgroup, since it clearly cannot contain the identity element. If $ba_i = e$, then $b = a_i^{-1}$, which implies $b \in A$, contrary to the assumption in Definition 10.7. In fact, bA contains no elements of A at all, for if $ba_i = a_j$, then $b = a_ja_i^{-1}$, and hence $b \in A$, again contrary to assumption.